

Discrete Sine Transform
The j, k entry is $\sin jk\pi h$

$$S = \begin{bmatrix} \sin \pi h & & \sin k\pi h \\ \vdots & \dots & \vdots & \dots \\ \sin n\pi h & & \sin nk\pi h \end{bmatrix}$$

Those eigenvectors are orthogonal just like the eigenfunctions: $\int_0^1 \sin j\pi x \sin k\pi x \, dx = 0$.

Problem Set 6.5

Problems 1–13 are about tests for positive definiteness.

- 1** Suppose the 2 by 2 tests $a > 0$ and $ac - b^2 > 0$ are passed. Then $c > b^2/a$ is also positive.

- (i) λ_1 and λ_2 have the *same sign* because their product $\lambda_1\lambda_2$ equals _____.
 (i) That sign is positive because $\lambda_1 + \lambda_2$ equals _____.

Conclusion: The tests $a > 0, ac - b^2 > 0$ guarantee positive eigenvalues λ_1, λ_2 .

- 2** Which of A_1, A_2, A_3, A_4 has two positive eigenvalues? Use the test, don't compute the λ 's. Find an $\mathbf{x}$ so that $\mathbf{x}^T A_1 \mathbf{x} < 0$, so A_1 fails the test.

$$A_1 = \begin{bmatrix} 5 & 6 \\ 6 & 7 \end{bmatrix} \quad A_2 = \begin{bmatrix} -1 & -2 \\ -2 & -5 \end{bmatrix} \quad A_3 = \begin{bmatrix} 1 & 10 \\ 10 & 100 \end{bmatrix} \quad A_4 = \begin{bmatrix} 1 & 10 \\ 10 & 101 \end{bmatrix}.$$

- 3** For which numbers b and c are these matrices positive definite?

$$A = \begin{bmatrix} 1 & b \\ b & 9 \end{bmatrix} \quad A = \begin{bmatrix} 2 & 4 \\ 4 & c \end{bmatrix} \quad A = \begin{bmatrix} c & b \\ b & c \end{bmatrix}.$$

With the pivots in D and multiplier in L , factor each A into LDL^T .

- 4** What is the quadratic $f = ax^2 + 2bxy + cy^2$ for each of these matrices? Complete the square to write f as a sum of one or two squares $d_1(\quad)^2 + d_2(\quad)^2$.

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 9 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 1 & 3 \\ 3 & 9 \end{bmatrix}.$$

- 5** Write $f(x, y) = x^2 + 4xy + 3y^2$ as a *difference* of squares and find a point (x, y) where f is negative. The minimum is not at $(0, 0)$ even though f has positive coefficients.
- 6** The function $f(x, y) = 2xy$ certainly has a saddle point and not a minimum at $(0, 0)$. What symmetric matrix A produces this f ? What are its eigenvalues?

- 7 Test to see if $R^T R$ is positive definite in each case:

$$R = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \quad \text{and} \quad R = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 2 & 1 \end{bmatrix} \quad \text{and} \quad R = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \end{bmatrix}.$$

- 8 The function $f(x, y) = 3(x + 2y)^2 + 4y^2$ is positive except at $(0, 0)$. What is the matrix in $f = [x \ y] A [x \ y]^T$? Check that the pivots of A are 3 and 4.
- 9 Find the 3 by 3 matrix A and its pivots, rank, eigenvalues, and determinant:

$$\begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} A \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 4(x_1 - x_2 + 2x_3)^2.$$

- 10 Which 3 by 3 symmetric matrices A and B produce these quadratics?

$$\mathbf{x}^T A \mathbf{x} = 2(x_1^2 + x_2^2 + x_3^2 - x_1 x_2 - x_2 x_3). \quad \text{Why is } A \text{ positive definite?}$$

$$\mathbf{x}^T B \mathbf{x} = 2(x_1^2 + x_2^2 + x_3^2 - x_1 x_2 - x_1 x_3 - x_2 x_3). \quad \text{Why is } B \text{ semidefinite?}$$

- 11 Compute the three upper left determinants of A to establish positive definiteness. Verify that their ratios give the second and third pivots.

$$\text{Pivots} = \text{ratios of determinants} \quad A = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 3 \\ 0 & 3 & 8 \end{bmatrix}.$$

- 12 For what numbers c and d are A and B positive definite? Test the 3 determinants:

$$A = \begin{bmatrix} c & 1 & 1 \\ 1 & c & 1 \\ 1 & 1 & c \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & d & 4 \\ 3 & 4 & 5 \end{bmatrix}.$$

- 13 Find a matrix with $a > 0$ and $c > 0$ and $a + c > 2b$ that has a negative eigenvalue.

Problems 14–20 are about applications of the tests.

- 14 If A is positive definite then A^{-1} is positive definite. Best proof: The eigenvalues of A^{-1} are positive because _____. Second proof (only for 2 by 2):

$$\text{The entries of } A^{-1} = \frac{1}{ac - b^2} \begin{bmatrix} c & -b \\ -b & a \end{bmatrix} \text{ pass the determinant tests } \underline{\hspace{2cm}}.$$

- 15 If A and B are positive definite, their sum $A + B$ is positive definite. Pivots and eigenvalues are not convenient for $A + B$. Better to prove $\mathbf{x}^T (A + B) \mathbf{x} > 0$. Or if $A = R^T R$ and $B = S^T S$, show that $A + B = [\mathbf{R} \ \mathbf{S}]^T \begin{bmatrix} \mathbf{R} \\ \mathbf{S} \end{bmatrix}$ with independent columns.

- 16 A positive definite matrix cannot have a zero (or even worse, a negative number) on its diagonal. Show that this matrix fails to have $\mathbf{x}^T A \mathbf{x} > 0$:

$$\begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 4 & 1 & 1 \\ 1 & 0 & 2 \\ 1 & 2 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \text{ is not positive when } (x_1, x_2, x_3) = (\quad, \quad, \quad).$$

- 17 A diagonal entry a_{jj} of a symmetric matrix cannot be smaller than all the λ 's. If it were, then $A - a_{jj}I$ would have _____ eigenvalues and would be positive definite. But $A - a_{jj}I$ has a _____ on the main diagonal.
- 18 If $A\mathbf{x} = \lambda\mathbf{x}$ then $\mathbf{x}^T A \mathbf{x} = ______$. If $\mathbf{x}^T A \mathbf{x} > 0$, prove that $\lambda > 0$.
- 19 Reverse Problem 18 to show that *if all $\lambda > 0$ then $\mathbf{x}^T A \mathbf{x} > 0$* . We must do this for *every* nonzero $\mathbf{x}$, not just the eigenvectors. So write $\mathbf{x}$ as a combination of the eigenvectors and *explain why all "cross terms" are $\mathbf{x}_i^T \mathbf{x}_j = 0$* . Then $\mathbf{x}^T A \mathbf{x}$ is
- $$(c_1\mathbf{x}_1 + \cdots + c_n\mathbf{x}_n)^T (c_1\lambda_1\mathbf{x}_1 + \cdots + c_n\lambda_n\mathbf{x}_n) = c_1^2\lambda_1\mathbf{x}_1^T\mathbf{x}_1 + \cdots + c_n^2\lambda_n\mathbf{x}_n^T\mathbf{x}_n > 0.$$
- 20 Give a quick reason why each of these statements is true:
- (a) Every positive definite matrix is invertible.
 - (b) The only positive definite projection matrix is $P = I$.
 - (c) A diagonal matrix with positive diagonal entries is positive definite.
 - (d) A symmetric matrix with a positive determinant might not be positive definite!

Problems 21–24 use the eigenvalues; Problems 25–27 are based on pivots.

- 21 For which s and t do A and B have all $\lambda > 0$ (therefore positive definite)?

$$A = \begin{bmatrix} s & -4 & -4 \\ -4 & s & -4 \\ -4 & -4 & s \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} t & 3 & 0 \\ 3 & t & 4 \\ 0 & 4 & t \end{bmatrix}.$$

- 22 From $A = Q\Lambda Q^T$ compute the positive definite symmetric square root $Q\Lambda^{1/2}Q^T$ of each matrix. Check that this square root gives $R^2 = A$:

$$A = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 10 & 6 \\ 6 & 10 \end{bmatrix}.$$

- 23 You may have seen the equation for an ellipse as $x^2/a^2 + y^2/b^2 = 1$. What are a and b when the equation is written $\lambda_1 x^2 + \lambda_2 y^2 = 1$? The ellipse $9x^2 + 4y^2 = 1$ has axes with half-lengths $a = ______$ and $b = ______$.
- 24 Draw the tilted ellipse $x^2 + xy + y^2 = 1$ and find the half-lengths of its axes from the eigenvalues of the corresponding matrix A .

- 25** With positive pivots in D , the factorization $A = LDL^T$ becomes $L\sqrt{D}\sqrt{D}L^T$. (Square roots of the pivots give $D = \sqrt{D}\sqrt{D}$.) Then $C = \sqrt{D}L^T$ yields the **Cholesky factorization** $A = C^TC$ which is “symmetrized LU ”:

$$\text{From } C = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} \text{ find } A. \quad \text{From } A = \begin{bmatrix} 4 & 8 \\ 8 & 25 \end{bmatrix} \text{ find } C = \text{chol}(A).$$

- 26** In the Cholesky factorization $A = C^TC$, with $C^T = L\sqrt{D}$, the square roots of the pivots are on the diagonal of C . Find C (upper triangular) for

$$A = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 2 & 8 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 7 \end{bmatrix}.$$

- 27** The symmetric factorization $A = LDL^T$ means that $x^T Ax = x^T LDL^T x$:

$$\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} a & b \\ b & c \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 1 & 0 \\ b/a & 1 \end{bmatrix} \begin{bmatrix} a & 0 \\ 0 & (ac - b^2)/a \end{bmatrix} \begin{bmatrix} 1 & b/a \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

The left side is $ax^2 + 2bxy + cy^2$. The right side is $a(x + \frac{b}{a}y)^2 + \text{_____}y^2$. The second pivot completes the square! Test with $a = 2, b = 4, c = 10$.

- 28** Without multiplying $A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$, find

- (a) the determinant of A (b) the eigenvalues of A
(c) the eigenvectors of A (d) a reason why A is symmetric positive definite.

- 29** For $F_1(x, y) = \frac{1}{4}x^4 + x^2y + y^2$ and $F_2(x, y) = x^3 + xy - x$ find the second derivative matrices H_1 and H_2 :

$$\text{Test for minimum: } H = \begin{bmatrix} \partial^2 F / \partial x^2 & \partial^2 F / \partial x \partial y \\ \partial^2 F / \partial y \partial x & \partial^2 F / \partial y^2 \end{bmatrix} \text{ is positive definite}$$

H_1 is positive definite so F_1 is concave up (= convex). Find the minimum point of F_1 and the saddle point of F_2 (look only where first derivatives are zero).

- 30** The graph of $z = x^2 + y^2$ is a bowl opening upward. The graph of $z = x^2 - y^2$ is a saddle. The graph of $z = -x^2 - y^2$ is a bowl opening downward. What is a test on a, b, c for $z = ax^2 + 2bxy + cy^2$ to have a saddle point at $(0, 0)$?
- 31** Which values of c give a bowl and which c give a saddle point for the graph of $z = 4x^2 + 12xy + cy^2$? Describe this graph at the borderline value of c .

Challenge Problems

- 32** A *group* of nonsingular matrices includes AB and A^{-1} if it includes A and B . “Products and inverses stay in the group.” Which of these are groups (as in 2.7.37)? Invent a “subgroup” of two of these groups (not I by itself = the smallest group).
- (a) Positive definite symmetric matrices A .
 - (b) Orthogonal matrices Q .
 - (c) All exponentials e^{tA} of a fixed matrix A .
 - (d) Matrices P with positive eigenvalues.
 - (e) Matrices D with determinant 1.
- 33** When A and B are symmetric positive definite, AB might not even be symmetric. But its eigenvalues are still positive. Start from $ABx = \lambda x$ and take dot products with Bx . Then prove $\lambda > 0$.
- 34** Write down the 5 by 5 sine matrix S from Worked Example 6.5 D, containing the eigenvectors of K when $n = 5$ and $h = 1/6$. Multiply K times S to see the five positive eigenvalues.
Their sum should equal the trace 10. Their product should be $\det K = 6$.
- 35** Suppose C is positive definite (so $y^T C y > 0$ whenever $y \neq \mathbf{0}$) and A has independent columns (so $Ax \neq \mathbf{0}$ whenever $x \neq \mathbf{0}$). Apply the energy test to $x^T A^T C A x$ to show that $A^T C A$ is positive definite: *the crucial matrix in engineering*.

6.6 Similar Matrices

The key step in this chapter is to diagonalize a matrix by using its eigenvectors. When S is the eigenvector matrix, the diagonal matrix $S^{-1}AS$ is Λ —the eigenvalue matrix. But diagonalization is not possible for every A . Some matrices have too few eigenvectors—we had to leave them alone. In this new section, the eigenvector matrix S remains the best choice when we can find it, *but now we allow any invertible matrix M* .

Starting from A we go to $M^{-1}AM$. This matrix may be diagonal—probably not. It still shares important properties of A . No matter which M we choose, *the eigenvalues stay the same*. The matrices A and $M^{-1}AM$ are called “similar”. A typical matrix A is similar to a whole family of other matrices because there are so many choices of M .

DEFINITION Let M be any invertible matrix. Then $B = M^{-1}AM$ is *similar* to A .

If $B = M^{-1}AM$ then immediately $A = MBM^{-1}$. That means: If B is similar to A then A is similar to B . The matrix in this reverse direction is M^{-1} —just as good as M .

A diagonalizable matrix is similar to Λ . In that special case M is S . We have $A = S\Lambda S^{-1}$ and $\Lambda = S^{-1}AS$. They certainly have the same eigenvalues! This section is opening up to other similar matrices $B = M^{-1}AM$, by allowing all invertible M .

The combination $M^{-1}AM$ appears when we change variables in a differential equation. Start with an equation for u and set $u = Mv$:

$$\frac{du}{dt} = Au \quad \text{becomes} \quad M \frac{dv}{dt} = AMv \quad \text{which is} \quad \frac{dv}{dt} = M^{-1}AMv.$$

The original coefficient matrix was A , the new one at the right is $M^{-1}AM$. Changing u to v leads to a similar matrix. When $M = S$ the new system is diagonal—the maximum in simplicity. Other choices of M could make the new system triangular and easier to solve. Since we can always go back to u , similar matrices must give the same growth or decay. More precisely, *the eigenvalues of A and B are the same*.

(No change in λ 's) Similar matrices A and $M^{-1}AM$ have the same eigenvalues. If x is an eigenvector of A , then $M^{-1}x$ is an eigenvector of $B = M^{-1}AM$.

The proof is quick, since $B = M^{-1}AM$ gives $A = MBM^{-1}$. Suppose $Ax = \lambda x$:

$$MBM^{-1}x = \lambda x \quad \text{means that} \quad B(M^{-1}x) = \lambda(M^{-1}x).$$

The eigenvalue of B is the same λ . The eigenvector has changed to $M^{-1}x$.

Two matrices can have the same *repeated* λ , and fail to be similar—as we will see.

Example 1 These matrices $M^{-1}AM$ all have the same eigenvalues 1 and 0.

The projection $A = \begin{bmatrix} .5 & .5 \\ .5 & .5 \end{bmatrix}$ is similar to $\Lambda = S^{-1}AS = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

Now choose $M = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}$. The similar matrix $M^{-1}AM$ is $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$.

Also choose $M = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. The similar matrix $M^{-1}AM$ is $\begin{bmatrix} .5 & -.5 \\ -.5 & .5 \end{bmatrix}$.

All 2 by 2 matrices with those eigenvalues 1 and 0 are similar to each other. The eigenvectors change with M , the eigenvalues don't change.

The eigenvalues in that example are *not repeated*. This makes life easy. Repeated eigenvalues are harder. The next example has eigenvalues 0 and 0. The zero matrix shares those eigenvalues, but it is similar only to itself: $M^{-1}0M = 0$.

Example 2 A family of similar matrices with $A = 0, 0$ (repeated eigenvalue)

$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ is similar to $\begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$ and all $B = \begin{bmatrix} cd & d^2 \\ -c^2 & -cd \end{bmatrix}$ except $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

These matrices B all have zero determinant (like A). They all have rank one (like A). One eigenvalue is zero and the trace is $cd - dc = 0$, so the other must be zero. I chose any $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ with $ad - bc = 1$, and $B = M^{-1}AM$.

These matrices B can't be diagonalized. In fact A is as close to diagonal as possible. It is the "**Jordan form**" for the family of matrices B . This is the outstanding member (my class says "Godfather") of the family. The Jordan form $J = A$ is as near as we can come to diagonalizing these matrices, when there is only one eigenvector. In going from A to $B = M^{-1}AM$, some things change and some don't. Here is a table to show this.

<u>Not changed by M</u>	<u>Changed by M</u>
Eigenvalues	Eigenvectors
Trace and determinant	Nullspace
Rank	Column space
Number of independent eigenvectors	Row space
Jordan form	Left nullspace
	Singular values

The eigenvalues don't change for similar matrices; the eigenvectors do. The trace is the sum of the λ 's (unchanged). The determinant is the product of the same λ 's.¹ The nullspace consists of the eigenvectors for $\lambda = 0$ (if any), so it can change. Its dimension $n - r$ does not change! The *number* of eigenvectors stays the same for each λ , while the vectors themselves are multiplied by M^{-1} . The *singular values* depend on $A^T A$, which definitely changes. They come in the next section.

¹The determinant is unchanged because $\det B = (\det M^{-1})(\det A)(\det M) = \det A$.

Examples of the Jordan Form

The *Jordan form* is the serious new idea here. We lead up to it with one more example of similar matrices: *triple eigenvalue, one eigenvector*.

Example 3 This **Jordan matrix** J has $\lambda = 5, 5, 5$ on its diagonal. Its only eigenvectors are multiples of $\mathbf{x} = (1, 0, 0)$. Algebraic multiplicity is 3, geometric multiplicity is 1:

$$\text{If } J = \begin{bmatrix} 5 & 1 & 0 \\ 0 & 5 & 1 \\ 0 & 0 & 5 \end{bmatrix} \text{ then } J - 5I = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \text{ has rank 2.}$$

Every similar matrix $B = M^{-1}JM$ has the same triple eigenvalue 5, 5, 5. Also $B - 5I$ must have the same rank 2. Its nullspace has dimension 1. So every B that is similar to this “Jordan block” J has only one independent eigenvector $M^{-1}\mathbf{x}$.

The transpose matrix J^T has the same eigenvalues 5, 5, 5, and $J^T - 5I$ has the same rank 2. **Jordan’s theorem says that J^T is similar to J .** The matrix M that produces the similarity happens to be the *reverse identity*:

$$J^T = M^{-1}JM \quad \text{is} \quad \begin{bmatrix} 5 & 0 & 0 \\ 1 & 5 & 0 \\ 0 & 1 & 5 \end{bmatrix} = \begin{bmatrix} & & 1 \\ & 1 & \\ 1 & & \end{bmatrix} \begin{bmatrix} 5 & 1 & 0 \\ 0 & 5 & 1 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix}.$$

All blank entries are zero. An eigenvector of J^T is $M^{-1}(1, 0, 0) = (0, 0, 1)$. There is one line of eigenvectors $(x_1, 0, 0)$ for J and another line $(0, 0, x_3)$ for J^T .

The key fact is that this matrix J is similar to *every* matrix A with eigenvalues 5, 5, 5 and *one line* of eigenvectors. There is an M with $M^{-1}AM = J$.

Example 4 Since J is as close to diagonal as we can get, the equation $du/dt = Ju$ cannot be simplified by changing variables. We must solve it as it stands:

$$\frac{du}{dt} = Ju = \begin{bmatrix} 5 & 1 & 0 \\ 0 & 5 & 1 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad \text{is} \quad \begin{aligned} dx/dt &= 5x + y \\ dy/dt &= 5y + z \\ dz/dt &= 5z. \end{aligned}$$

The system is triangular. We think naturally of back substitution. Solve the last equation and work upwards. Main point: *All solutions contain e^{5t} since $\lambda = 5$:*

$$\begin{aligned} \text{Last equation} \quad \frac{dz}{dt} &= 5z && \text{yields} \quad z = z(0)e^{5t} \\ \text{Notice } te^{5t} \quad \frac{dy}{dt} &= 5y + z && \text{yields} \quad y = (y(0) + tz(0))e^{5t} \\ \text{Notice } t^2e^{5t} \quad \frac{dx}{dt} &= 5x + y && \text{yields} \quad x = (x(0) + ty(0) + \tfrac{1}{2}t^2z(0))e^{5t}. \end{aligned}$$

The two missing eigenvectors are responsible for the te^{5t} and t^2e^{5t} terms in y and x . The factors t and t^2 enter because $\lambda = 5$ is a triple eigenvalue with one eigenvector.

Note Chapter 7 will explain another approach to similar matrices. Instead of changing variables by $u = Mv$, we “*change the basis*”. In this approach, similar matrices will represent the same transformation of n -dimensional space. When we choose a basis for $\mathbf{R}^n$, we get a matrix. The standard basis vectors ($M = I$) lead to $I^{-1}AI$ which is A . Other bases lead to similar matrices $B = M^{-1}AM$.

The Jordan Form

For every A , we want to choose M so that $M^{-1}AM$ is as *nearly diagonal as possible*. When A has a full set of n eigenvectors, they go into the columns of M . Then $M = S$. The matrix $S^{-1}AS$ is diagonal, period. This matrix Λ is the Jordan form of A —when A can be diagonalized. In the general case, eigenvectors are missing and Λ can’t be reached.

Suppose A has s independent eigenvectors. Then it is similar to a matrix with s blocks. Each block is like J in Example 3. *The eigenvalue is on the diagonal with 1’s just above it.* This block accounts for one eigenvector of A . When there are n eigenvectors and n blocks, they are all 1 by 1. In that case J is Λ .

(Jordan form) If A has s independent eigenvectors, it is similar to a matrix J that has s Jordan blocks on its diagonal: Some matrix M puts A into Jordan form:

Jordan form
$$M^{-1}AM = \begin{bmatrix} J_1 & & \\ & \ddots & \\ & & J_s \end{bmatrix} = J. \quad (1)$$

Each block in J has one eigenvalue λ_i , one eigenvector, and 1’s above the diagonal:

Jordan block
$$J_i = \begin{bmatrix} \lambda_i & 1 & & \\ & \cdot & \cdot & \\ & & \cdot & 1 \\ & & & \lambda_i \end{bmatrix}. \quad (2)$$

A is similar to B if they share the same Jordan form J —not otherwise.

The Jordan form J has an off-diagonal 1 for each missing eigenvector (and the 1’s are next to the eigenvalues). This is the big theorem about matrix similarity. In every family of similar matrices, we are picking one outstanding member called J . It is nearly diagonal (or if possible completely diagonal). For that J , we can solve $du/dt = Ju$ as in Example 4. We can take powers J^k as in Problems 9–10. Every other matrix in the family has the form $A = MJM^{-1}$. The connection through M solves $du/dt = Au$.

The point you must see is that $MJM^{-1}MJM^{-1} = MJ^2M^{-1}$. That cancellation of $M^{-1}M$ in the middle has been used through this chapter (when M was S). We found A^{100} from $S\Lambda^{100}S^{-1}$ —by diagonalizing the matrix. Now we can’t quite diagonalize A . So we use $MJ^{100}M^{-1}$ instead.

Jordan's Theorem is proved in my textbook *Linear Algebra and Its Applications*. Please refer to that book (or more advanced books) for the proof. The reasoning is rather intricate and in actual computations the Jordan form is not at all popular—its calculation is not stable. A slight change in A will separate the repeated eigenvalues and remove the off-diagonal 1's—switching to a diagonal Λ .

Proved or not, you have caught the central idea of similarity—to make A as simple as possible while preserving its essential properties.

■ REVIEW OF THE KEY IDEAS ■

1. B is similar to A if $B = M^{-1}AM$, for some invertible matrix M .
2. Similar matrices have the same eigenvalues. Eigenvectors are multiplied by M^{-1} .
3. If A has n independent eigenvectors then A is similar to Λ (take $M = S$).
4. Every matrix is similar to a Jordan matrix J (which has Λ as its diagonal part). J has a block for each eigenvector, and 1's for missing eigenvectors.

■ WORKED EXAMPLES ■

6.6 A The 4 by 4 triangular Pascal matrix A and its inverse (alternating diagonals) are

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 1 & 3 & 3 & 1 \end{bmatrix} \quad \text{and} \quad A^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ -1 & 3 & -3 & 1 \end{bmatrix}.$$

Check that A and A^{-1} have the same eigenvalues. Find a diagonal matrix D with alternating signs that gives $A^{-1} = D^{-1}AD$. This A is similar to A^{-1} , which is unusual.

These similar matrices must have the same Jordan form J . This J has only one block because the Pascal matrix has only one line of eigenvectors.

Solution The triangular matrices A and A^{-1} both have $\lambda = 1, 1, 1, 1$ on their main diagonals. Choose D with alternating 1 and -1 on its diagonal. D equals D^{-1} :

$$D^{-1}AD = \begin{bmatrix} -1 & & & \\ & 1 & & \\ & & -1 & \\ & & & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 1 & 3 & 3 & 1 \end{bmatrix} \begin{bmatrix} -1 & & & \\ & 1 & & \\ & & -1 & \\ & & & 1 \end{bmatrix} = A^{-1}.$$

Check: Changing signs in rows 1 and 3 of A , and columns 1 and 3, produces the four negative entries in A^{-1} . We are multiplying row i by $(-1)^i$ and column j by $(-1)^j$, which gives the alternating diagonals in A^{-1} . Then AD has *columns with alternating signs*.

6.6 B *The best way to compute eigenvalues of a large matrix is not from solving $\det(A - \lambda I) = 0$. That high degree polynomial is a numerical disaster.*

Instead we compute similar matrices $A_1, A_2, \dots$ that approach a triangular matrix. Then the eigenvalues of A (unchanged) are almost sitting on the main diagonal.

One way is to factor $A = QR$ by “Gram-Schmidt”. Reverse the order to $A_1 = RQ$. This matrix is similar to A because $RQ = Q^{-1}(QR)Q$. An example with $c = \cos \theta$ and $s = \sin \theta$ shows how a small off-diagonal s can be *cubed* in A_1 :

$$A = \begin{bmatrix} c & s \\ s & 0 \end{bmatrix} \text{ factors into } \begin{bmatrix} c & s \\ s & -c \end{bmatrix} \begin{bmatrix} 1 & cs \\ 0 & s^2 \end{bmatrix} = QR.$$

$$A_1 = RQ = \begin{bmatrix} c + cs^2 & s^3 \\ s^3 & -cs^2 \end{bmatrix} \text{ has } s^3 \text{ below the diagonal}$$

Another step can factor $A_1 = Q_1 R_1$ and reverse to $A_2 = R_1 Q_1$. This **QR method** is in Section 9.3 with a further improvement for A_1 . Add cs^2 to its diagonal (to get zero in the corner) and then subtract back from A_2 :

$$\text{Shift and factor } A_1 + cs^2 I = Q_1 R_1 \quad \text{Reverse and shift back } A_2 = R_1 Q_1 - cs^2 I$$

Shifted QR is an amazing success—just about the best way to compute eigenvalues.

Problem Set 6.6

- 1 If $C = F^{-1}AF$ and also $C = G^{-1}BG$, what matrix M gives $B = M^{-1}AM$?
Conclusion: If C is similar to A and also to B then ____.
- 2 If $A = \text{diag}(1, 3)$ and $B = \text{diag}(3, 1)$ show that A and B are similar (find an M).
- 3 Show that A and B are similar by finding M so that $B = M^{-1}AM$:

$$\begin{array}{ll} A = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} & \text{and} \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \\ A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} & \text{and} \quad B = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \\ A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} & \text{and} \quad B = \begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix}. \end{array}$$

- 4 If a 2 by 2 matrix A has eigenvalues 0 and 1, why is it similar to $\Lambda = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$?
Deduce from Problem 1 that all 2 by 2 matrices with those eigenvalues are similar.
- 5 Which of these six matrices are similar? Check their eigenvalues.

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}.$$

- 6 There are sixteen 2 by 2 matrices whose entries are 0's and 1's. Similar matrices go into the same family. How many families? How many matrices (total 16) in each family?
- 7 (a) If x is in the nullspace of A show that $M^{-1}x$ is in the nullspace of $M^{-1}AM$.
 (b) The nullspaces of A and $M^{-1}AM$ have the same (vectors)(basis)(dimension).
- 8 Suppose $Ax = \lambda x$ and $Bx = \lambda x$ with the same λ 's and x 's. With n independent eigenvectors we have $A = B$: Why? Find $A \neq B$ when both have eigenvalues 0, 0 but only one line of eigenvectors $(x_1, 0)$.
- 9 By direct multiplication find A^2 and A^3 and A^5 when

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

Guess the form of A^k . Set $k = 0$ to find A^0 and $k = -1$ to find A^{-1} .

Questions 10–14 are about the Jordan form.

- 10 By direct multiplication, find J^2 and J^3 when

$$J = \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix}.$$

Guess the form of J^k . Set $k = 0$ to find J^0 . Set $k = -1$ to find J^{-1} .

- 11 Solve $du/dt = Ju$ for J in Problem 10, starting from $u(0) = (5, 2)$. Remember $te^{\lambda t}$.
- 12 These Jordan matrices have eigenvalues 0, 0, 0, 0. They have two eigenvectors (one from each block). But the block sizes don't match and they are *not similar*:

$$J = \left[\begin{array}{cc|cc} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right] \quad \text{and} \quad K = \left[\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 \end{array} \right].$$

For any matrix M , compare JM with MK . If they are equal show that M is not invertible. Then $M^{-1}JM = K$ is impossible: J is *not similar* to K .

- 13 Based on Problem 12, what are the five Jordan forms when $\lambda = 0, 0, 0, 0$?
- 14 Prove that A^T is always similar to A (we know the λ 's are the same):
1. For one Jordan block J_i : Find M_i so that $M_i^{-1}J_iM_i = J_i^T$ (see Example 3).
 2. For any J with blocks J_i : Build M_0 from blocks so that $M_0^{-1}JM_0 = J^T$.
 3. For any $A = MJM^{-1}$: Show that A^T is similar to J^T and so to J and to A .

- 15 Prove that $\det(A - \lambda I) = \det(M^{-1}AM - \lambda I)$. (You could write $I = M^{-1}M$ and factor out $\det M^{-1}$ and $\det M$.) Since these *characteristic polynomials* of A and $M^{-1}AM$ are the same, the eigenvalues are the same (with the same multiplicities).

- 16 Which pairs are similar? Choose a, b, c, d to prove that the other pairs aren't:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \begin{bmatrix} b & a \\ d & c \end{bmatrix} \quad \begin{bmatrix} c & d \\ a & b \end{bmatrix} \quad \begin{bmatrix} d & c \\ b & a \end{bmatrix}.$$

- 17 True or false, with a good reason:

- (a) A symmetric matrix can't be similar to a nonsymmetric matrix.
- (b) An invertible matrix can't be similar to a singular matrix.
- (c) A can't be similar to $-A$ unless $A = 0$.
- (d) A can't be similar to $A + I$.

- 18 If B is invertible, prove that AB is similar to BA . *They have the same eigenvalues.*

- 19 If A is 6 by 4 and B is 4 by 6, AB and BA have different sizes. But with blocks

$$M^{-1}FM = \begin{bmatrix} I & -A \\ 0 & I \end{bmatrix} \begin{bmatrix} AB & 0 \\ B & 0 \end{bmatrix} \begin{bmatrix} I & A \\ 0 & I \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ B & BA \end{bmatrix} = G.$$

- (a) What sizes are the four blocks (the same four sizes in each matrix)?
 - (b) This equation is $M^{-1}FM = G$, so F and G have the same 10 eigenvalues. F has the 6 eigenvalues of AB plus 4 zeros; G has the 4 eigenvalues of BA plus 6 zeros. **AB has the same eigenvalues as BA plus _____ zeros.**
- 20 Why are these statements all true?
- (a) If A is similar to B then A^2 is similar to B^2 .
 - (b) A^2 and B^2 can be similar when A and B are not similar (try $\lambda = 0, 0$).
 - (c) $\begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}$ is similar to $\begin{bmatrix} 3 & 1 \\ 0 & 4 \end{bmatrix}$.
 - (d) $\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$ is not similar to $\begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$.
 - (e) If we exchange rows 1 and 2 of A , and then exchange columns 1 and 2, **the eigenvalues stay the same.** In this case $M = \underline{\hspace{2cm}}$.
- 21 If J is the 5 by 5 Jordan block with $\lambda = 0$, find J^2 and count its eigenvectors and find its Jordan form (there will be two blocks).

Challenge Problems

- 22 If an n by n matrix A has all eigenvalues $\lambda = 0$, prove that $A^n =$ zero matrix. (Maybe prove first that $J^n =$ zero matrix, by direct multiplication. Or use the Cayley-Hamilton Theorem?)
- 23 For the shifted QR method in the Worked Example 6.6 B, show that A_2 is similar to A_1 . No change in eigenvalues, and the A 's quickly approach a diagonal matrix.
- 24 If A is similar to A^{-1} , must all the eigenvalues equal 1 or -1 ?

6.7 Singular Value Decomposition (SVD)

The Singular Value Decomposition is a highlight of linear algebra. A is any m by n matrix, square or rectangular. Its rank is r . We will diagonalize this A , but not by $S^{-1}AS$. The eigenvectors in S have three big problems: They are usually not orthogonal, there are not always enough eigenvectors, and $Ax = \lambda x$ requires A to be square. The *singular vectors* of A solve all those problems in a perfect way.

The price we pay is to have two sets of singular vectors, u 's and v 's. The u 's are eigenvectors of AA^T and the v 's are eigenvectors of A^TA . Since those matrices are both symmetric, their eigenvectors can be chosen orthonormal. In equation (13) below, the simple fact that A times A^TA is the same as AA^T times A will lead to a remarkable property of these u 's and v 's:

$$\text{"A is diagonalized"} \quad Av_1 = \sigma_1 u_1 \quad Av_2 = \sigma_2 u_2 \quad \dots \quad Av_r = \sigma_r u_r \quad (1)$$

The singular vectors $v_1, \dots, v_r$ are in the *row space* of A . The outputs $u_1, \dots, u_r$ are in the *column space* of A . The *singular values* $\sigma_1, \dots, \sigma_r$ are all positive numbers. When the v 's and u 's go into the columns of V and U , orthogonality gives $V^TV = I$ and $U^TU = I$. The σ 's go into a diagonal matrix Σ .

Just as $Ax_i = \lambda_i x_i$ led to the diagonalization $AS = S\Lambda$, the equations $Av_i = \sigma_i u_i$ tell us column by column that $AV = U\Sigma$:

$$\begin{array}{c} (m \text{ by } n)(n \text{ by } r) \\ \text{equals} \\ (m \text{ by } r)(r \text{ by } r) \end{array} \quad A \begin{bmatrix} v_1 & \cdots & v_r \end{bmatrix} = \begin{bmatrix} u_1 & \cdots & u_r \end{bmatrix} \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix}. \quad (2)$$

This is the heart of the SVD, but there is more. Those v 's and u 's account for the row space and column space of A . We need $n - r$ more v 's and $m - r$ more u 's, from the nullspace $N(A)$ and the left nullspace $N(A^T)$. They can be orthonormal bases for those two nullspaces (and then automatically orthogonal to the first r v 's and u 's). Include all the v 's and u 's in V and U , so these matrices become *square*. *We still have* $AV = U\Sigma$.

$$\begin{array}{c} (m \text{ by } n)(n \text{ by } n) \\ \text{equals} \\ (m \text{ by } m)(m \text{ by } n) \end{array} \quad A \begin{bmatrix} v_1 & \cdots & v_r & \cdots & v_n \end{bmatrix} = \begin{bmatrix} u_1 & \cdots & u_r & \cdots & u_m \end{bmatrix} \begin{bmatrix} \sigma_1 & & & & \\ & \ddots & & & \\ & & \sigma_r & & \\ & & & \ddots & \\ & & & & 0 \end{bmatrix} \quad (3)$$

The new Σ is m by n . It is just the old r by r matrix (call that Σ_r) with $m - r$ new zero rows and $n - r$ new zero columns. The real change is in the shapes of U and V and Σ . Still $V^TV = I$ and $U^TU = I$, with sizes n and m .

V is now a square orthogonal matrix, with inverse $V^{-1} = V^T$. So $AV = U\Sigma$ can become $A = U\Sigma V^T$. This is the *Singular Value Decomposition*:

$$\text{SVD} \quad A = U\Sigma V^T = u_1\sigma_1v_1^T + \cdots + u_r\sigma_rv_r^T. \quad (4)$$

I would write the earlier “reduced SVD” from equation (2) as $A = U_r \Sigma_r V_r^T$. That is equally true, without the extra zeros in Σ . This reduced SVD gives the same splitting of A into a sum of r matrices, each of rank one.

We will see that $\sigma_i^2 = \lambda_i$ is an eigenvalue of $A^T A$ and also AA^T . When we put the singular values in descending order, $\sigma_1 \geq \sigma_2 \geq \dots \sigma_r > 0$, the splitting in equation (4) gives the r rank-one pieces of A *in order of importance*.

Example 1 When is $U \Sigma V^T$ (singular values) the same as $S \Lambda S^{-1}$ (eigenvalues)?

Solution We need orthonormal eigenvectors in $S = U$. We need nonnegative eigenvalues in $\Lambda = \Sigma$. So A must be a *positive semidefinite (or definite) symmetric matrix* $Q \Lambda Q^T$.

Example 2 If $A = xy^T$ with unit vectors x and y , what is the SVD of A ?

Solution The reduced SVD in (2) is exactly xy^T , with rank $r = 1$. It has $u_1 = x$ and $v_1 = y$ and $\sigma_1 = 1$. For the full SVD, complete $u_1 = x$ to an orthonormal basis of u 's, and complete $v_1 = y$ to an orthonormal basis of v 's. No new σ 's.

I will describe an application before proving that $Av_i = \sigma_i u_i$. This key equation gave the diagonalizations (2) and (3) and (4) of the SVD: $A = U \Sigma V^T$.

Image Compression

Unusually, I am going to stop the theory and describe applications. This is the century of data, and often that data is stored in a matrix. *A digital image is really a matrix of pixel values*. Each little picture element or “pixel” has a gray scale number between black and white (it has three numbers for a color picture). The picture might have $512 = 2^9$ pixels in each row and $256 = 2^8$ pixels down each column. We have a 256 by 512 pixel matrix with 2^{17} entries! To store one picture, the computer has no problem. But a CT or MR scan produces an image at every cross section—a ton of data. If the pictures are frames in a movie, 30 frames a second means 108,000 images per hour. Compression is especially needed for high definition digital TV, or the equipment could not keep up in real time.

What is compression? We want to replace those 2^{17} matrix entries by a smaller number, *without losing picture quality*. A simple way would be to use larger pixels—replace groups of four pixels by their average value. This is 4 : 1 compression. But if we carry it further, like 16 : 1, our image becomes “blocky”. We want to replace the mn entries by a smaller number, in a way that the human visual system won't notice.

Compression is a billion dollar problem and everyone has ideas. Later in this book I will describe Fourier transforms (used in **jpeg**) and wavelets (now in **JPEG2000**). Here we try an SVD approach: *Replace the 256 by 512 pixel matrix by a matrix of rank one: a column times a row*. If this is successful, the storage requirement becomes $256 + 512$ (add instead of multiply). The compression ratio $(256)(512)/(256 + 512)$ is better than 170 to 1. This is more than we hope for. We may actually use five matrices of rank one (so a matrix approximation of rank 5). The compression is still 34 : 1 and the crucial question is the picture quality.

Where does the SVD come in? *The best rank one approximation to A is the matrix $\sigma_1 \mathbf{u}_1 \mathbf{v}_1^T$.* It uses the largest singular value σ_1 . The best rank 5 approximation includes also $\sigma_2 \mathbf{u}_2 \mathbf{v}_2^T + \cdots + \sigma_5 \mathbf{u}_5 \mathbf{v}_5^T$. **The SVD puts the pieces of A in descending order.**

A library compresses a different matrix. The rows correspond to key words. Columns correspond to titles in the library. The entry in this *word-title matrix* is $a_{ij} = 1$ if word i is in title j (otherwise $a_{ij} = 0$). We normalize the columns so long titles don't get an advantage. We might use a table of contents or an abstract. (Other books might share the title "*Introduction to Linear Algebra*".) Instead of $a_{ij} = 1$, the entries of A can include the *frequency* of the search words. See Section 8.6 for the SVD in statistics.

Once the indexing matrix is created, the search is a linear algebra problem. This giant matrix has to be compressed. The SVD approach gives an optimal low rank approximation, better for library matrices than for natural images. There is an ever-present tradeoff in the cost to compute the $\mathbf{u}$'s and $\mathbf{v}$'s. We still need a better way (with sparse matrices).

The Bases and the SVD

Start with a 2 by 2 matrix. Let its rank be $r = 2$, so A is invertible. We want $\mathbf{v}_1$ and $\mathbf{v}_2$ to be perpendicular unit vectors. **We also want $A\mathbf{v}_1$ and $A\mathbf{v}_2$ to be perpendicular.** (This is the tricky part. It is what makes the bases special.) Then the unit vectors $\mathbf{u}_1 = A\mathbf{v}_1 / \|A\mathbf{v}_1\|$ and $\mathbf{u}_2 = A\mathbf{v}_2 / \|A\mathbf{v}_2\|$ will be orthonormal. Here is a specific example:

$$\text{Unsymmetric matrix} \quad A = \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix}. \quad (5)$$

No orthogonal matrix Q will make $Q^{-1}AQ$ diagonal. We need $U^{-1}AV$. The two bases will be different—one basis cannot do it. The output is $A\mathbf{v}_1 = \sigma_1 \mathbf{u}_1$ when the input is $\mathbf{v}_1$. The "singular values" σ_1 and σ_2 are the lengths $\|A\mathbf{v}_1\|$ and $\|A\mathbf{v}_2\|$.

$$\begin{aligned} AV &= U\Sigma \\ A &= U\Sigma V^T \end{aligned} \quad A \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 \end{bmatrix} = \begin{bmatrix} \sigma_1 \mathbf{u}_1 & \sigma_2 \mathbf{u}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 \end{bmatrix} \begin{bmatrix} \sigma_1 & \\ & \sigma_2 \end{bmatrix}. \quad (6)$$

There is a neat way to remove U and see V by itself. Multiply A^T times A .

$$A^T A = (U\Sigma V^T)^T (U\Sigma V^T) = V\Sigma^T \Sigma V^T. \quad (7)$$

$U^T U$ disappears because it equals I . (We require $\mathbf{u}_1^T \mathbf{u}_1 = 1 = \mathbf{u}_2^T \mathbf{u}_2$ and $\mathbf{u}_1^T \mathbf{u}_2 = 0$.) Multiplying those diagonal Σ^T and Σ gives σ_1^2 and σ_2^2 . That leaves an ordinary diagonalization of the crucial symmetric matrix $A^T A$, whose eigenvalues are σ_1^2 and σ_2^2 :

$$\begin{array}{l} \text{Eigenvalues } \sigma_1^2, \sigma_2^2 \\ \text{Eigenvectors } \mathbf{v}_1, \mathbf{v}_2 \end{array} \quad A^T A = V \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix} V^T. \quad (8)$$

This is exactly like $A = Q\Lambda Q^T$. But the symmetric matrix is not A itself. Now the symmetric matrix is $A^T A$! And the columns of V are the eigenvectors of $A^T A$. Last is U :

Compute the eigenvectors v and eigenvalues σ^2 of $A^T A$. Then each $u = Av/\sigma$.

For large matrices LAPACK finds a special way to avoid multiplying $A^T A$ in $\text{svd}(A)$.

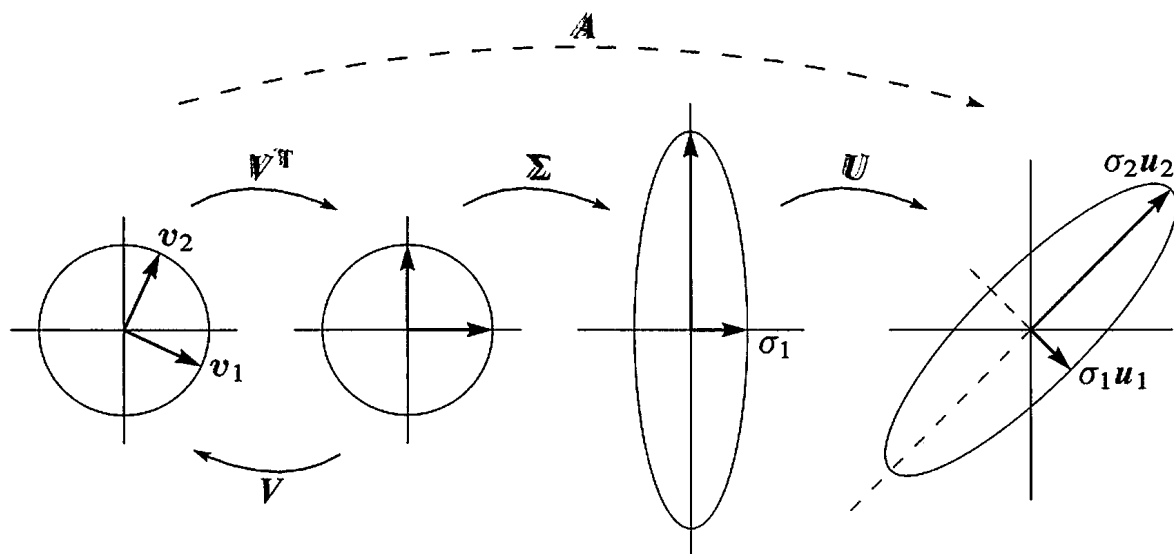


Figure 6.8: U and V are rotations and reflections. Σ stretches circle to ellipse.

Example 3 Find the singular value decomposition of that matrix $A = \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix}$.

Solution Compute $A^T A$ and its eigenvectors. Then make them unit vectors:

$$A^T A = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix} \text{ has unit eigenvectors } v_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \text{ and } v_2 = \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}.$$

The eigenvalues of $A^T A$ are 8 and 2. The v 's are perpendicular, because eigenvectors of every symmetric matrix are perpendicular—and $A^T A$ is automatically symmetric.

Now the u 's are quick to find, because Av_1 is going to be in the direction of u_1 :

$$Av_1 = \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 2\sqrt{2} \\ 0 \end{bmatrix}. \text{ The unit vector is } u_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Clearly Av_1 is the same as $2\sqrt{2} u_1$. The first singular value is $\sigma_1 = 2\sqrt{2}$. Then $\sigma_1^2 = 8$.

$$Av_2 = \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 0 \\ \sqrt{2} \end{bmatrix}. \text{ The unit vector is } u_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Now Av_2 is $\sqrt{2} u_2$ and $\sigma_2 = \sqrt{2}$. Thus σ_2^2 agrees with the other eigenvalue 2 of $A^T A$.

$$A = U\Sigma V^T \text{ is } \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2\sqrt{2} & \\ & \sqrt{2} \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}. \quad (9)$$

This matrix, and every invertible 2 by 2 matrix, *transforms the unit circle to an ellipse*. You can see that in the figure, which was created by Cliff Long and Tom Hern.

One final point about that example. We found the u 's from the v 's. Could we find the u 's directly? Yes, by multiplying AA^T instead of $A^T A$:

$$\text{Use } V^T V = I \quad AA^T = (U \Sigma V^T)(V \Sigma^T U^T) = U \Sigma \Sigma^T U^T. \quad (10)$$

Multiplying $\Sigma \Sigma^T$ gives σ_1^2 and σ_2^2 as before. *The u 's are eigenvectors of AA^T :*

$$\text{Diagonal in this example} \quad AA^T = \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 8 & 0 \\ 0 & 2 \end{bmatrix}.$$

The eigenvectors $(1, 0)$ and $(0, 1)$ agree with u_1 and u_2 found earlier. Why take the first eigenvector to be $(1, 0)$ instead of $(-1, 0)$ or $(0, 1)$? Because we have to follow Av_1 (I missed that in my video lecture ...). Notice that AA^T has the same eigenvalues (8 and 2) as $A^T A$. The singular values are $\sqrt{8}$ and $\sqrt{2}$.

Example 4 Find the SVD of the singular matrix $A = \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}$. The rank is $r = 1$.

Solution The row space has only one basis vector $v_1 = (1, 1)/\sqrt{2}$. The column space has only one basis vector $u_1 = (2, 1)/\sqrt{5}$. Then $Av_1 = (4, 2)/\sqrt{2}$ must equal $\sigma_1 u_1$. It does, with $\sigma_1 = \sqrt{10}$.

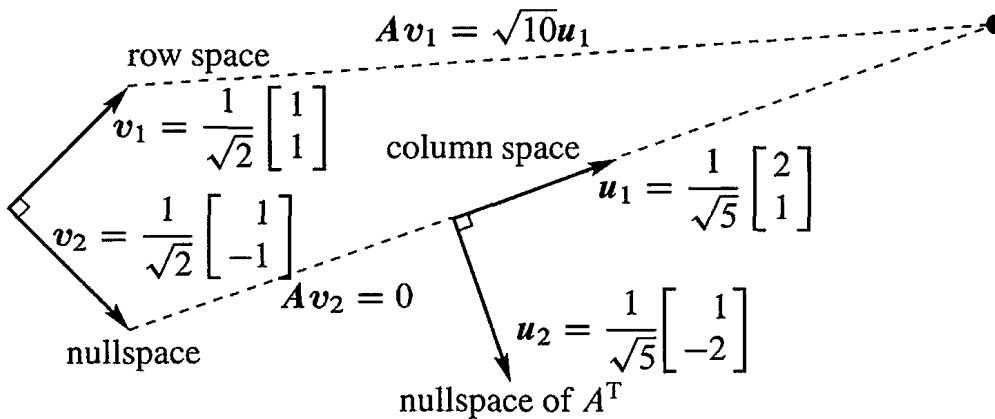


Figure 6.9: The SVD chooses orthonormal bases for 4 subspaces so that $Av_i = \sigma_i u_i$.

The SVD could stop after the row space and column space (it usually doesn't). It is customary for U and V to be square. The matrices need a second column. **The vector v_2 is in the nullspace.** It is perpendicular to v_1 in the row space. Multiply by A to get $Av_2 = 0$. We could say that the second singular value is $\sigma_2 = 0$, but singular values are like pivots—only the r nonzeros are counted.

$$\text{Full size} \quad A = U \Sigma V^T \quad \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} \sqrt{10} & 0 \\ 0 & 0 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}. \quad (11)$$

The matrices U and V contain orthonormal bases for all four subspaces:

first	r	columns of V :	row space of A
last	$n - r$	columns of V :	nullspace of A
first	r	columns of U :	column space of A
last	$m - r$	columns of U :	nullspace of A^T

The first columns $v_1, \dots, v_r$ and $u_1, \dots, u_r$ are eigenvectors of $A^T A$ and $A A^T$. We now explain why $A v_i$ falls in the direction of u_i . The last v 's and u 's (in the nullspaces) are easier. As long as those are orthonormal, the SVD will be correct.

Proof of the SVD: Start from $A^T A v_i = \sigma_i^2 v_i$, which gives the v 's and σ 's. Multiplying by v_i^T leads to $\|A v_i\|^2$. To prove that $A v_i = \sigma_i u_i$, the key step is to multiply by A :

$$v_i^T A^T A v_i = \sigma_i^2 v_i^T v_i \quad \text{gives} \quad \|A v_i\|^2 = \sigma_i^2 \quad \text{so that} \quad \|A v_i\| = \sigma_i \quad (12)$$

$$A A^T A v_i = \sigma_i^2 A v_i \quad \text{gives} \quad u_i = A v_i / \sigma_i \quad \text{as a unit eigenvector of } A A^T. \quad (13)$$

Equation (12) used the small trick of placing parentheses in $(v_i^T A^T)(A v_i) = \|A v_i\|^2$. Equation (13) placed the all-important parentheses in $(A A^T)(A v_i)$. This shows that $A v_i$ is an eigenvector of $A A^T$. Divide by its length σ_i to get the unit vector $u_i = A v_i / \sigma_i$. These u 's are orthogonal because $(A v_i)^T (A v_j) = v_i^T (A^T A v_j) = v_i^T (\sigma_j^2 v_j) = 0$.

I will give my opinion directly. The SVD is the climax of this linear algebra course. I think of it as the final step in the Fundamental Theorem. First come the *dimensions* of the four subspaces. Then their *orthogonality*. Then *the orthonormal bases diagonalize A* . It is all in the formula $A = U \Sigma V^T$. You have made it to the top.

Eigshow (Part 2)

Section 6.1 described the MATLAB demo called **eigshow**. The first option is *eig*, when x moves in a circle and Ax follows on an ellipse. The second option is *svd*, when two vectors x and y stay perpendicular as they travel around a circle. Then Ax and Ay move too (not usually perpendicular). The four vectors are in the Java demo on web.mit.edu/18.06.

The SVD is seen graphically when Ax is perpendicular to Ay . Their directions at that moment give an orthonormal basis u_1, u_2 . Their lengths give the singular values σ_1, σ_2 . The vectors x and y at that same moment are the orthonormal basis v_1, v_2 .

Searching the Web

I will end with an application of the SVD to web search engines. When you google a word, you get a list of web sites in order of importance. You could try “four subspaces”.

The HITS algorithm that we describe is one way to produce that ranked list. It begins with about 200 sites found from an index of key words, and after that we look only at links between pages. Search engines are link-based more than content-based.

Start with the 200 sites and all sites that link to them and all sites they link to. That is our list, to be put in order. Importance can be measured by links out and links in.

1. The site is an **authority**: *links come in from many sites*. Especially from hubs.
2. The site is a **hub**: *links go out to many sites in the list*. Especially to authorities.

We want numbers $x_1, \dots, x_N$ to rank the authorities and $y_1, \dots, y_N$ to rank the hubs. Start with a simple count: x_i^0 and y_i^0 count the links into and out of site i .

Here is the point: *A good authority has links from important sites* (like hubs). Links from universities count more heavily than links from friends. *A good hub is linked to important sites* (like authorities). A link to **amazon.com** unfortunately means more than a link to **wellesleycambridge.com**. The rankings x^0 and y^0 from counting links are updated to x^1 and y^1 by taking account of *good* links (measuring their quality by x^0 and y^0):

$$\text{Authority values } x_i^1 = \sum_{j \text{ links to } i} y_j^0 \qquad \text{Hub values } y_i^1 = \sum_{i \text{ links to } j} x_j^0 \quad (14)$$

In matrix language those are $x^1 = A^T y^0$ and $y^1 = A x^0$. The matrix A contains 1's and 0's, with $a_{ij} = 1$ when i links to j . In the language of graphs, A is an “adjacency matrix” for the World Wide Web (an enormous matrix). The new x^1 and y^1 give better rankings, but not the best. Take another step like (14), to reach x^2 and y^2 :

$$A^T A \text{ and } A A^T \text{ appear } x^2 = A^T y^1 = A^T A x^0 \quad \text{and} \quad y^2 = A^T x^1 = A A^T y^0. \quad (15)$$

In two steps we are multiplying by $A^T A$ and $A A^T$. Twenty steps will multiply by $(A^T A)^{10}$ and $(A A^T)^{10}$. When we take powers, the largest eigenvalue σ_1^2 begins to dominate. And the vectors x and y line up with the leading eigenvectors v_1 and u_1 of $A^T A$ and $A A^T$. We are computing the top terms in the SVD, by the *power method* that is discussed in Section 9.3. It is wonderful that linear algebra helps to understand the Web.

Google actually creates rankings by a random walk that follows web links. The more often this random walk goes to a site, the higher the ranking. The frequency of visits gives the leading eigenvector ($\lambda = 1$) of the normalized adjacency matrix for the Web. ***That Markov matrix has 2.7 billion rows and columns, from 2.7 billion web sites.***

This is the largest eigenvalue problem ever solved. The excellent book by Langville and Meyer, *Google's PageRank and Beyond*, explains in detail the science of search engines. See mathworks.com/company/newsletter/clevescorner/oct02_cleve.shtml

But many of the important techniques are well-kept secrets of Google. Probably Google starts with last month's eigenvector as a first approximation, and runs the random walk very fast. To get a high ranking, you want a lot of links from important sites. The HITS algorithm is described in the 1999 *Scientific American* (June 16). But I don't think the SVD is mentioned there. . .

■ REVIEW OF THE KEY IDEAS ■

1. The SVD factors A into $U \Sigma V^T$, with r singular values $\sigma_1 \geq \dots \geq \sigma_r > 0$.

2. The numbers $\sigma_1^2, \dots, \sigma_r^2$ are the nonzero eigenvalues of AA^T and $A^T A$.
3. The orthonormal columns of U and V are eigenvectors of AA^T and $A^T A$.
4. Those columns hold orthonormal bases for the four fundamental subspaces of A .
5. Those bases diagonalize the matrix: $Av_i = \sigma_i u_i$ for $i \leq r$. This is $AV = U\Sigma$.

■ WORKED EXAMPLES ■

6.7 A Identify by name these decompositions $A = c_1 b_1 + \dots + c_r b_r$ of an m by n matrix. Each term is a rank one matrix (column c times row b). The rank of A is r .

1. *Orthogonal* columns $c_1, \dots, c_r$ and *orthogonal* rows $b_1, \dots, b_r$.
2. *Orthogonal* columns $c_1, \dots, c_r$ and *triangular* rows $b_1, \dots, b_r$.
3. *Triangular* columns $c_1, \dots, c_r$ and *triangular* rows $b_1, \dots, b_r$.

$A = CB$ is $(m \text{ by } r)(r \text{ by } n)$. Triangular vectors c_i and b_i have zeros up to component i . The matrix C with columns c_i is lower triangular, the matrix B with rows b_i is upper triangular. Where do the rank and the pivots and singular values come into this picture?

Solution These three splittings $A = CB$ are basic to linear algebra, pure or applied:

1. **Singular Value Decomposition** $A = U\Sigma V^T$ (*orthogonal* U , *orthogonal* ΣV^T)
2. **Gram-Schmidt Orthogonalization** $A = QR$ (*orthogonal* Q , *triangular* R)
3. **Gaussian Elimination** $A = LU$ (*triangular* L , *triangular* U)

You might prefer to separate out the σ_i and pivots d_i and heights h_i :

1. $A = U\Sigma V^T$ with unit vectors in U and V . The singular values are in Σ .
2. $A = QHR$ with unit vectors in Q and diagonal 1's in R . The heights h_i are in H .
3. $A = LDU$ with diagonal 1's in L and U . The pivots are in D .

Each h_i tells the height of column i above the base from earlier columns. The volume of the full n -dimensional box ($r = m = n$) comes from $A = U\Sigma V^T = LDU = QHR$:

$$|\det A| = |\text{product of } \sigma \text{'s}| = |\text{product of } d \text{'s}| = |\text{product of } h \text{'s}|.$$

6.7.B For $A = xy^T$ of rank one (2 by 2), compare $A = U\Sigma V^T$ with $A = SAS^{-1}$.

Comment This started as an exam problem in 2007. It led further and became interesting. Now there is an essay called "The Four Fundamental Subspaces: 4 Lines" on web.mit.edu/18.06. The Jordan form enters when $y^T x = 0$ and $\lambda = 0$ is repeated.

6.7.C Show that $\sigma_1 \geq |\lambda|_{\max}$. The largest singular value dominates all eigenvalues.
 Show that $\sigma_1 \geq |a_{ij}|_{\max}$. The largest singular value dominates all entries of A .

Solution Start from $A = U\Sigma V^T$. Remember that multiplying by an orthogonal matrix *does not change length*: $\|Qx\| = \|x\|$ because $\|Qx\|^2 = x^T Q^T Q x = x^T x = \|x\|^2$. This applies to $Q = U$ and $Q = V^T$. In between is the diagonal matrix Σ .

$$\|Ax\| = \|U\Sigma V^T x\| = \|\Sigma V^T x\| \leq \sigma_1 \|V^T x\| = \sigma_1 \|x\|. \quad (16)$$

An eigenvector has $\|Ax\| = |\lambda| \|x\|$. So (16) says that $|\lambda| \|x\| \leq \sigma_1 \|x\|$. Then $|\lambda| \leq \sigma_1$.

Apply also to the unit vector $x = (1, 0, \dots, 0)$. Now Ax is the first column of A . Then by inequality (16), this column has length $\leq \sigma_1$. Every entry must have magnitude $\leq \sigma_1$.

Example 5 Estimate the singular values σ_1 and σ_2 of A and A^{-1} :

$$\text{Eigenvalues} = 1 \quad A = \begin{bmatrix} 1 & 0 \\ C & 1 \end{bmatrix} \quad \text{and} \quad A^{-1} = \begin{bmatrix} 1 & 0 \\ -C & 1 \end{bmatrix}. \quad (17)$$

Solution The length of the first column is $\sqrt{1 + C^2} \leq \sigma_1$, from the reasoning above. This confirms that $\sigma_1 \geq 1$ and $\sigma_1 \geq C$. Then σ_1 dominates the eigenvalues 1, 1 and the entry C . If C is very large then σ_1 is much bigger than the eigenvalues.

This matrix A has determinant = 1. $A^T A$ also has determinant = 1 and then $\sigma_1 \sigma_2 = 1$. For this matrix, $\sigma_1 \geq 1$ and $\sigma_1 \geq C$ lead to $\sigma_2 \leq 1$ and $\sigma_2 \leq 1/C$.

Conclusion: If $C = 1000$ then $\sigma_1 \geq 1000$ and $\sigma_2 \leq 1/1000$. A is *ill-conditioned*, slightly sick. Inverting A is easy by algebra, but solving $Ax = b$ by elimination could be dangerous. A is close to a singular matrix even though both eigenvalues are $\lambda = 1$. By slightly changing the 1, 2 entry from zero to $1/C = 1/1000$, the matrix becomes singular.

Section 9.2 will explain how the ratio $\sigma_{\max}/\sigma_{\min}$ governs the roundoff error in elimination. MATLAB warns you if this “condition number” is large. Here $\sigma_1/\sigma_2 \geq C^2$.

Problem Set 6.7

Problems 1–3 compute the SVD of a square singular matrix A .

1 Find the eigenvalues and unit eigenvectors v_1, v_2 of $A^T A$. Then find $u_1 = Av_1/\sigma_1$:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \quad \text{and} \quad A^T A = \begin{bmatrix} 10 & 20 \\ 20 & 40 \end{bmatrix} \quad \text{and} \quad AA^T = \begin{bmatrix} 5 & 15 \\ 15 & 45 \end{bmatrix}.$$

Verify that u_1 is a unit eigenvector of AA^T . Complete the matrices U, Σ, V .

$$\text{SVD} \quad \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} = \begin{bmatrix} u_1 & u_2 \end{bmatrix} \begin{bmatrix} \sigma_1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 & v_2 \end{bmatrix}^T.$$

- 2 Write down orthonormal bases for the four fundamental subspaces of this A .
- 3 (a) Why is the trace of $A^T A$ equal to the sum of all a_{ij}^2 ?
 (b) For every rank-one matrix, why is $\sigma_1^2 = \text{sum of all } a_{ij}^2$?

Problems 4–7 ask for the SVD of matrices of rank 2.

- 4 Find the eigenvalues and unit eigenvectors of $A^T A$ and $A A^T$. Keep each $A v = \sigma u$:

Fibonacci matrix $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$

Construct the singular value decomposition and verify that A equals $U \Sigma V^T$.

- 5 Use the `svd` part of the MATLAB demo `eigshow` to find those v 's graphically.
- 6 Compute $A^T A$ and $A A^T$ and their eigenvalues and unit eigenvectors for V and U .

Rectangular matrix $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$.

Check $AV = U \Sigma$ (this will decide $\pm$ signs in U). Σ has the same shape as A .

- 7 What is the closest rank-one approximation to that 2 by 3 matrix?
- 8 A square invertible matrix has $A^{-1} = V \Sigma^{-1} U^T$. This says that the *singular values of A^{-1}* are $1/\sigma(A)$. Show that $\sigma_{\max}(A^{-1}) \sigma_{\max}(A) \geq 1$.
- 9 Suppose $u_1, \dots, u_n$ and $v_1, \dots, v_n$ are orthonormal bases for $\mathbf{R}^n$. Construct the matrix A that transforms each v_j into u_j to give $A v_1 = u_1, \dots, A v_n = u_n$.
- 10 Construct the matrix with rank one that has $A v = 12u$ for $v = \frac{1}{2}(1, 1, 1, 1)$ and $u = \frac{1}{3}(2, 2, 1)$. Its only singular value is $\sigma_1 = \underline{\hspace{1cm}}$.
- 11 Suppose A has orthogonal columns $w_1, w_2, \dots, w_n$ of lengths $\sigma_1, \sigma_2, \dots, \sigma_n$. What are U , Σ , and V in the SVD?
- 12 Suppose A is a 2 by 2 symmetric matrix with unit eigenvectors u_1 and u_2 . If its eigenvalues are $\lambda_1 = 3$ and $\lambda_2 = -2$, what are the matrices U , Σ , V^T in its SVD?
- 13 If $A = QR$ with an orthogonal matrix Q , the SVD of A is almost the same as the SVD of R . Which of the three matrices U , Σ , V is changed because of Q ?
- 14 Suppose A is invertible (with $\sigma_1 > \sigma_2 > 0$). Change A by *as small a matrix as possible* to produce a singular matrix A_0 . Hint: U and V do not change:

From $A = \begin{bmatrix} u_1 & u_2 \end{bmatrix} \begin{bmatrix} \sigma_1 & \\ & \sigma_2 \end{bmatrix} \begin{bmatrix} v_1 & v_2 \end{bmatrix}^T$ find the nearest A_0 .

- 15 Why doesn't the SVD for $A + I$ just use $\Sigma + I$?

Challenge Problems

- 16 (Search engine) Run a **random walk** $x(2), \dots, x(n)$ starting from web site $x(1) = 1$. Count the visits to each site. At each step the code chooses the next website $x(k)$ with probabilities given by column $x(k-1)$ of A . At the end, p gives the fraction of time at each site from a histogram: count visits. **The rankings are based on p .**

Please compare p to the steady state eigenvector of the Markov matrix A :

$$A = [0 \ .1 \ .2 \ .7; \ .05 \ 0 \ .15 \ .8; \ .15 \ .25 \ 0 \ .6; \ .1 \ .3 \ .6 \ 0]'$$

$n = 100$; $x = \text{zeros}(1, n)$; $x(1) = 1$;

for $k = 2 : n$ $x(k) = \min(\text{find}(\text{rand} < \text{cumsum}(A(:, x(k-1)))))$; end

$p = \text{hist}(x, 1 : 4)/n$

- 17 The $1, -1$ first difference matrix A has $A^T A =$ second difference matrix. The singular vectors of A are *sine vectors* v and *cosine vectors* u . Then $Av = \sigma u$ is the discrete form of $d/dx(\sin cx) = c(\cos cx)$. This is the best SVD I have seen.

$$\text{SVD of } A \quad A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \quad A^T A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

$$\text{Orthogonal sine matrix} \quad V = \frac{1}{\sqrt{2}} \begin{bmatrix} \sin \pi/4 & \sin 2\pi/4 & \sin 3\pi/4 \\ \sin 2\pi/4 & \sin 4\pi/4 & \sin 6\pi/4 \\ \sin 3\pi/4 & \sin 6\pi/4 & \sin 9\pi/4 \end{bmatrix}$$

- (a) Put numbers in V : The unit eigenvectors of $A^T A$ are singular vectors of A . Show that the columns of V have $A^T A v = \lambda v$ with $\lambda = 2 - \sqrt{2}, 2, 2 + \sqrt{2}$.
- (b) Multiply AV and verify that its columns are orthogonal. They are $\sigma_1 u_1$ and $\sigma_2 u_2$ and $\sigma_3 u_3$. The first columns of the cosine matrix U are u_1, u_2, u_3 .
- (c) Since A is 4 by 3, we need a fourth orthogonal vector u_4 . It comes from the nullspace of A^T . What is u_4 ?

The cosine vectors in U are eigenvectors of AA^T . The fourth cosine is $(1, 1, 1, 1)/2$.

$$AA^T = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \quad U = \frac{1}{\sqrt{2}} \begin{bmatrix} \cos \pi/8 & \cos 2\pi/8 & \cos 3\pi/8 \\ \cos 3\pi/8 & \cos 6\pi/8 & \cos 9\pi/8 \\ \cos 5\pi/8 & \cos 10\pi/8 & \cos 15\pi/8 \\ \cos 7\pi/8 & \cos 14\pi/8 & \cos 21\pi/8 \end{bmatrix}$$

Those angles $\pi/8, 3\pi/8, 5\pi/8, 7\pi/8$ fit 4 points with spacing $\pi/4$ between 0 and π . The sine transform has three points $\pi/4, 2\pi/4, 3\pi/4$. The full cosine transform includes u_4 from the "zero frequency" or *direct current* eigenvector $(1, 1, 1, 1)$.

The 8 by 8 cosine transform in 2D is the workhorse of **jpeg compression**. Linear algebra (circulant, Toeplitz, orthogonal matrices) is at the heart of signal processing.

Table of Eigenvalues and Eigenvectors

How are the properties of a matrix reflected in its eigenvalues and eigenvectors? This question is fundamental throughout Chapter 6. A table that organizes the key facts may be helpful. Here are the special properties of the eigenvalues λ_i and the eigenvectors x_i .

Symmetric: $A^T = A$	real λ 's	orthogonal $x_i^T x_j = 0$
Orthogonal: $Q^T = Q^{-1}$	all $ \lambda = 1$	orthogonal $\bar{x}_i^T x_j = 0$
Skew-symmetric: $A^T = -A$	imaginary λ 's	orthogonal $\bar{x}_i^T x_j = 0$
Complex Hermitian: $\bar{A}^T = A$	real λ 's	orthogonal $\bar{x}_i^T x_j = 0$
Positive Definite: $x^T A x > 0$	all $\lambda > 0$	orthogonal since $A^T = A$
Markov: $m_{ij} > 0, \sum_{i=1}^n m_{ij} = 1$	$\lambda_{\max} = 1$	steady state $x > 0$
Similar: $B = M^{-1} A M$	$\lambda(B) = \lambda(A)$	$x(B) = M^{-1} x(A)$
Projection: $P = P^2 = P^T$	$\lambda = 1; 0$	column space; nullspace
Plane Rotation	$e^{i\theta}$ and $e^{-i\theta}$	$x = (1, i)$ and $(1, -i)$
Reflection: $I - 2uu^T$	$\lambda = -1; 1, \dots, 1$	u ; whole plane $u^\perp$
Rank One: uv^T	$\lambda = v^T u; 0, \dots, 0$	u ; whole plane $v^\perp$
Inverse: A^{-1}	$1/\lambda(A)$	keep eigenvectors of A
Shift: $A + cI$	$\lambda(A) + c$	keep eigenvectors of A
Stable Powers: $A^n \rightarrow 0$	all $ \lambda < 1$	any eigenvectors
Stable Exponential: $e^{At} \rightarrow 0$	all $\text{Re } \lambda < 0$	any eigenvectors
Cyclic Permutation: row 1 of I last	$\lambda_k = e^{2\pi i k/n}$	$x_k = (1, \lambda_k, \dots, \lambda_k^{n-1})$
Tridiagonal: $-1, 2, -1$ on diagonals	$\lambda_k = 2 - 2 \cos \frac{k\pi}{n+1}$	$x_k = \left(\sin \frac{k\pi}{n+1}, \sin \frac{2k\pi}{n+1}, \dots \right)$
Diagonalizable: $A = S \Lambda S^{-1}$	diagonal of Λ	columns of S are independent
Symmetric: $A = Q \Lambda Q^T$	diagonal of Λ (real)	columns of Q are orthonormal
Schur: $A = Q T Q^{-1}$	diagonal of T	columns of Q if $A^T A = A A^T$
Jordan: $J = M^{-1} A M$	diagonal of J	each block gives $x = (0, \dots, 1, \dots, 0)$
Rectangular: $A = U \Sigma V^T$	$\text{rank}(A) = \text{rank}(\Sigma)$	eigenvectors of $A^T A, A A^T$ in V, U

Chapter 7

Linear Transformations

7.1 The Idea of a Linear Transformation

When a matrix A multiplies a vector v , it “transforms” v into another vector Av . *In goes v , out comes $T(v) = Av$.* A transformation T follows the same idea as a function. In goes a number x , out comes $f(x)$. For one vector v or one number x , we multiply by the matrix or we evaluate the function. The deeper goal is to see all v ’s at once. We are transforming the whole space V when we multiply every v by A .

Start again with a matrix A . It transforms v to Av . It transforms w to Aw . Then we know what happens to $u = v + w$. There is no doubt about Au , it has to equal $Av + Aw$. Matrix multiplication $T(v) = Av$ gives a *linear transformation*:

A *transformation* T assigns an output $T(v)$ to each input vector v in V . The transformation is *linear* if it meets these requirements for all v and w :

$$(a) \ T(v + w) = T(v) + T(w) \qquad (b) \ T(cv) = cT(v) \quad \text{for all } c.$$

If the input is $v = 0$, the output must be $T(v) = 0$. We combine (a) and (b) into one:

$$\text{Linear transformation} \quad T(cv + dw) \text{ must equal } cT(v) + dT(w).$$

Again I can test matrix multiplication for linearity: $A(cv + dw) = cAv + dAw$ is true.

A linear transformation is highly restricted. Suppose T adds u_0 to every vector. Then $T(v) = v + u_0$ and $T(w) = w + u_0$. This isn’t good, or at least *it isn’t linear*. Applying T to $v + w$ produces $v + w + u_0$. That is not the same as $T(v) + T(w)$:

$$\text{Shift is not linear} \quad v + w + u_0 \text{ is not } T(v) + T(w) = v + u_0 + w + u_0.$$

The exception is when $u_0 = 0$. The transformation reduces to $T(v) = v$. This is the *identity transformation* (nothing moves, as in multiplication by the identity matrix). That is certainly linear. In this case the input space V is the same as the output space W .

The linear-plus-shift transformation $T(v) = Av + u_0$ is called “*affine*”. Straight lines stay straight although T is not linear. Computer graphics works with affine transformations in Section 8.6, because we must be able to move images.

Example 1 Choose a fixed vector $a = (1, 3, 4)$, and let $T(v)$ be the dot product $a \cdot v$:

The input is $v = (v_1, v_2, v_3)$. The output is $T(v) = a \cdot v = v_1 + 3v_2 + 4v_3$.

This is linear. The inputs v come from three-dimensional space, so $V = \mathbf{R}^3$. The outputs are just numbers, so the output space is $W = \mathbf{R}^1$. We are multiplying by the row matrix $A = [1 \ 3 \ 4]$. Then $T(v) = Av$.

You will get good at recognizing which transformations are linear. If the output involves squares or products or lengths, v_1^2 or $v_1 v_2$ or $\|v\|$, then T is not linear.

Example 2 The length $T(v) = \|v\|$ is not linear. Requirement (a) for linearity would be $\|v + w\| = \|v\| + \|w\|$. Requirement (b) would be $\|cv\| = c\|v\|$. Both are false!

Not (a): The sides of a triangle satisfy an *inequality* $\|v + w\| \leq \|v\| + \|w\|$.

Not (b): The length $\|-v\|$ is not $-\|v\|$. For negative c , we fail.

Example 3 (Important) T is the transformation that *rotates every vector* by 30° . The “*domain*” is the xy plane (all input vectors v). The “*range*” is also the xy plane (all rotated vectors $T(v)$). We described T without a matrix: rotate by 30° .

Is rotation linear? *Yes it is.* We can rotate two vectors and add the results. The sum of rotations $T(v) + T(w)$ is the same as the rotation $T(v + w)$ of the sum. The whole plane is turning together, in this linear transformation.

Lines to Lines, Triangles to Triangles

Figure 7.1 shows the line from v to w in the input space. It also shows the line from $T(v)$ to $T(w)$ in the output space. Linearity tells us: Every point on the input line goes onto the output line. And more than that: *Equally spaced points go to equally spaced points.* The middle point $u = \frac{1}{2}v + \frac{1}{2}w$ goes to the middle point $T(u) = \frac{1}{2}T(v) + \frac{1}{2}T(w)$.

The second figure moves up a dimension. Now we have three corners v_1, v_2, v_3 . Those inputs have three outputs $T(v_1), T(v_2), T(v_3)$. *The input triangle goes onto the output triangle.* Equally spaced points stay equally spaced (along the edges, and then between the edges). The middle point $u = \frac{1}{3}(v_1 + v_2 + v_3)$ goes to the middle point $T(u) = \frac{1}{3}(T(v_1) + T(v_2) + T(v_3))$.

The rule of linearity extends to combinations of three vectors or n vectors:

$$\begin{array}{ll} \text{Linearity} & u = c_1 v_1 + c_2 v_2 + \cdots + c_n v_n \quad \text{transforms to} \\ & T(u) = c_1 T(v_1) + c_2 T(v_2) + \cdots + c_n T(v_n) \end{array} \quad (1)$$

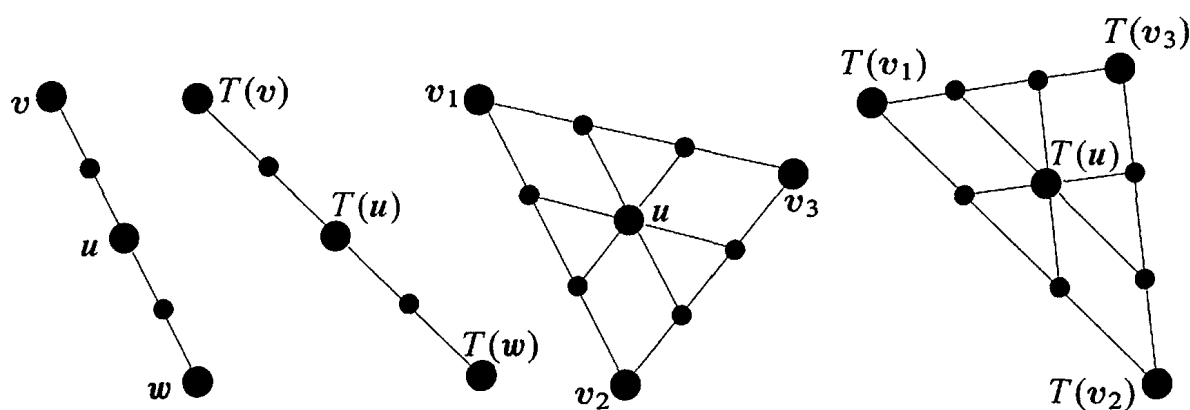


Figure 7.1: Lines to lines, equal spacing to equal spacing, $u = \mathbf{0}$ to $T(u) = \mathbf{0}$.

Note Transformations have a language of their own. Where there is no matrix, we can't talk about a column space. But the idea can be rescued. The column space consisted of all outputs Av . The nullspace consisted of all inputs for which $Av = \mathbf{0}$. Translate those into “range” and “kernel”:

Range of T = set of all outputs $T(v)$: range corresponds to column space

Kernel of T = set of all inputs for which $T(v) = \mathbf{0}$: kernel corresponds to nullspace.

The range is in the output space W . The kernel is in the input space V . When T is multiplication by a matrix, $T(v) = Av$, you can translate to column space and nullspace.

Examples of Transformations (mostly linear)

Example 4 Project every 3-dimensional vector straight down onto the xy plane. Then $T(x, y, z) = (x, y, 0)$. The range is that plane, which contains every $T(v)$. The kernel is the z axis (which projects down to zero). This projection is linear.

Example 5 Project every 3-dimensional vector onto the horizontal plane $z = 1$. The vector $v = (x, y, z)$ is transformed to $T(v) = (x, y, 1)$. This transformation is not linear. Why not? It doesn't even transform $v = \mathbf{0}$ into $T(v) = \mathbf{0}$.

Multiply every 3-dimensional vector by a 3 by 3 matrix A . This $T(v) = Av$ is linear.

$$T(v + w) = A(v + w) \quad \text{does equal} \quad Av + Aw = T(v) + T(w).$$

Example 6 Suppose A is an invertible matrix. The kernel of T is the zero vector; the range W equals the domain V . Another linear transformation is multiplication by A^{-1} . This is the **inverse transformation** T^{-1} , which brings every vector $T(v)$ back to v :

$$T^{-1}(T(v)) = v \quad \text{matches the matrix multiplication} \quad A^{-1}(Av) = v.$$

We are reaching an unavoidable question. **Are all linear transformations from $V = \mathbf{R}^n$ to $W = \mathbf{R}^m$ produced by matrices?** When a linear T is described as a “rotation” or “projection” or “. . .”, is there always a matrix hiding behind T ?

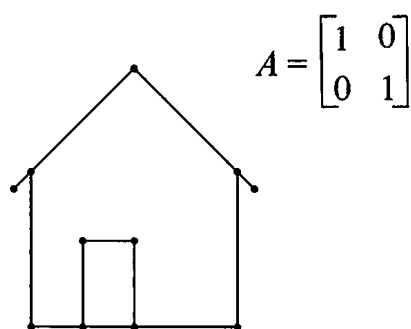
The answer is yes. This is an approach to linear algebra that doesn't start with matrices. The next section shows that we still end up with matrices.

Linear Transformations of the Plane

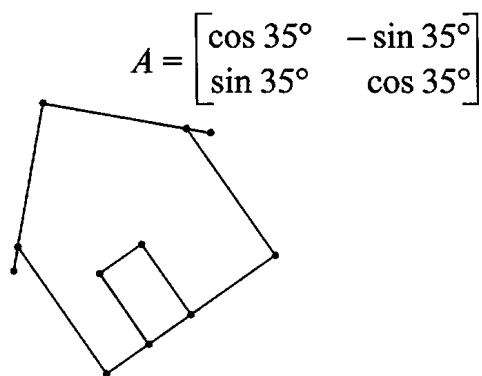
It is more interesting to *see* a transformation than to define it. When a 2 by 2 matrix A multiplies all vectors in $\mathbf{R}^2$, we can watch how it acts. Start with a “house” that has eleven endpoints. Those eleven vectors v are transformed into eleven vectors Av . Straight lines between v ’s become straight lines between the transformed vectors Av . (The transformation from house to house is linear!) Applying A to a standard house produces a new house—possibly stretched or rotated or otherwise unlivable.

This part of the book is visual, not theoretical. We will show four houses and the matrices that produce them. The columns of H are the eleven corners of the first house. (H is 2 by 12, so **plot2d** will connect the 11th corner to the first.) The 11 points in the house matrix H are multiplied by A to produce the corners AH of the other houses.

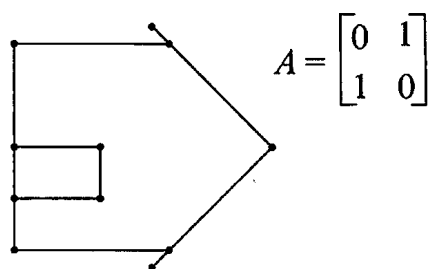
$$\text{House matrix } H = \begin{bmatrix} -6 & -6 & -7 & 0 & 7 & 6 & 6 & -3 & -3 & 0 & 0 & -6 \\ -7 & 2 & 1 & 8 & 1 & 2 & -7 & -7 & -2 & -2 & -7 & -7 \end{bmatrix}.$$



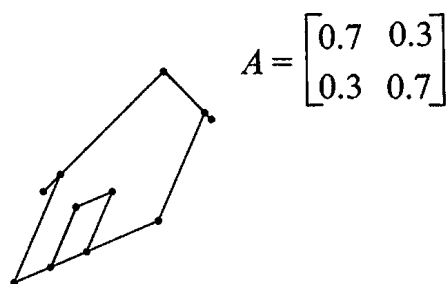
$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$



$$A = \begin{bmatrix} \cos 35^\circ & -\sin 35^\circ \\ \sin 35^\circ & \cos 35^\circ \end{bmatrix}$$



$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$



$$A = \begin{bmatrix} 0.7 & 0.3 \\ 0.3 & 0.7 \end{bmatrix}$$

Figure 7.2: Linear transformations of a house drawn by **plot2d**($A * H$).

■ REVIEW OF THE KEY IDEAS ■

1. A transformation T takes each v in the input space to $T(v)$ in the output space.
2. T is **linear** if $T(v + w) = T(v) + T(w)$ and $T(cv) = cT(v)$: lines to lines.

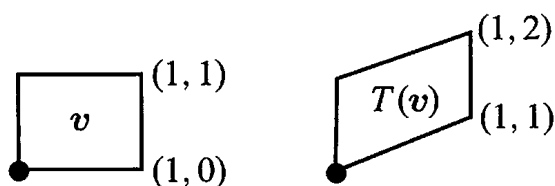
3. Combinations to combinations: $T(c_1 v_1 + \cdots + c_n v_n) = c_1 T(v_1) + \cdots + c_n T(v_n)$.
4. The transformation $T(v) = Av + v_0$ is linear only if $v_0 = \mathbf{0}$. Then $T(v) = Av$.

■ WORKED EXAMPLES ■

7.1 A The elimination matrix $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ gives a *shearing transformation* from (x, y) to $T(x, y) = (x, x + y)$. Draw the xy plane and show what happens to $(1, 0)$ and $(1, 1)$. What happens to points on the vertical lines $x = 0$ and $x = a$? If the inputs fill the unit square $0 \leq x \leq 1$, $0 \leq y \leq 1$, draw the outputs (the transformed square).

Solution The points $(1, 0)$ and $(2, 0)$ on the x axis transform by T to $(1, 1)$ and $(2, 2)$. The horizontal x axis transforms to the 45° line (going through $(0, 0)$ of course). The points on the y axis are *not moved* because $T(0, y) = (0, y)$. The y axis is the line of eigenvectors of T with $\lambda = 1$. Points with $x = a$ move up by a .

Vertical lines slide up
This is the shearing
Squares to parallelograms



7.1 B A **nonlinear transformation** T is invertible if every $\mathbf{b}$ in the output space comes from exactly one $\mathbf{x}$ in the input space: $T(\mathbf{x}) = \mathbf{b}$ always has exactly one solution. Which of these transformations (on real numbers x) is invertible and what is T^{-1} ? *None are linear, not even T_3 .* When you solve $T(x) = \mathbf{b}$, you are inverting T :

$$T_1(x) = x^2 \quad T_2(x) = x^3 \quad T_3(x) = x + 9 \quad T_4(x) = e^x \quad T_5(x) = \frac{1}{x} \quad \text{for nonzero } x\text{'s}$$

Solution T_1 is not invertible: $x^2 = 1$ has *two* solutions and $x^2 = -1$ has *no* solution. T_4 is not invertible because $e^x = -1$ has no solution. (If the output space changes to *positive* b 's then the inverse of $e^x = b$ is $x = \ln b$.)

Notice $T_5^2 = \text{identity}$. But $T_3^2(x) = x + 18$. What are $T_2^2(x)$ and $T_4^2(x)$?

T_2, T_3, T_5 are invertible. The solutions to $x^3 = b$ and $x + 9 = b$ and $\frac{1}{x} = b$ are unique:

$$x = T_2^{-1}(b) = b^{1/3} \quad x = T_3^{-1}(b) = b - 9 \quad x = T_5^{-1}(b) = 1/b$$

Problem Set 7.1

- 1 A linear transformation must leave the zero vector fixed: $T(\mathbf{0}) = \mathbf{0}$. Prove this from $T(\mathbf{v} + \mathbf{w}) = T(\mathbf{v}) + T(\mathbf{w})$ by choosing $\mathbf{w} = \underline{\hspace{2cm}}$ (and finish the proof). Prove it also from $T(c\mathbf{v}) = cT(\mathbf{v})$ by choosing $c = \underline{\hspace{2cm}}$.
- 2 Requirement (b) gives $T(c\mathbf{v}) = cT(\mathbf{v})$ and also $T(d\mathbf{w}) = dT(\mathbf{w})$. Then by addition, requirement (a) gives $T(\underline{\hspace{1cm}}) = (\underline{\hspace{1cm}})$. What is $T(c\mathbf{v} + d\mathbf{w} + e\mathbf{u})$?
- 3 Which of these transformations are not linear? The input is $\mathbf{v} = (v_1, v_2)$:
 - (a) $T(\mathbf{v}) = (v_2, v_1)$ (b) $T(\mathbf{v}) = (v_1, v_1)$ (c) $T(\mathbf{v}) = (0, v_1)$
 - (d) $T(\mathbf{v}) = (0, 1)$ (e) $T(\mathbf{v}) = v_1 - v_2$ (f) $T(\mathbf{v}) = v_1 v_2$.
- 4 If S and T are linear transformations, is $S(T(\mathbf{v}))$ linear or quadratic?
 - (a) (Special case) If $S(\mathbf{v}) = \mathbf{v}$ and $T(\mathbf{v}) = \mathbf{v}$, then $S(T(\mathbf{v})) = \mathbf{v}$ or $\mathbf{v}^2$?
 - (b) (General case) $S(\mathbf{w}_1 + \mathbf{w}_2) = S(\mathbf{w}_1) + S(\mathbf{w}_2)$ and $T(\mathbf{v}_1 + \mathbf{v}_2) = T(\mathbf{v}_1) + T(\mathbf{v}_2)$ combine into

$$S(T(\mathbf{v}_1 + \mathbf{v}_2)) = S(\underline{\hspace{2cm}}) = \underline{\hspace{2cm}} + \underline{\hspace{2cm}}.$$
- 5 Suppose $T(\mathbf{v}) = \mathbf{v}$ except that $T(0, v_2) = (0, 0)$. Show that this transformation satisfies $T(c\mathbf{v}) = cT(\mathbf{v})$ but not $T(\mathbf{v} + \mathbf{w}) = T(\mathbf{v}) + T(\mathbf{w})$.
- 6 Which of these transformations satisfy $T(\mathbf{v} + \mathbf{w}) = T(\mathbf{v}) + T(\mathbf{w})$ and which satisfy $T(c\mathbf{v}) = cT(\mathbf{v})$?
 - (a) $T(\mathbf{v}) = \mathbf{v}/\|\mathbf{v}\|$ (b) $T(\mathbf{v}) = v_1 + v_2 + v_3$ (c) $T(\mathbf{v}) = (v_1, 2v_2, 3v_3)$
 - (d) $T(\mathbf{v}) = \text{largest component of } \mathbf{v}$.
- 7 For these transformations of $\mathbf{V} = \mathbf{R}^2$ to $\mathbf{W} = \mathbf{R}^2$, find $T(T(\mathbf{v}))$. Is this transformation T^2 linear?
 - (a) $T(\mathbf{v}) = -\mathbf{v}$ (b) $T(\mathbf{v}) = \mathbf{v} + (1, 1)$
 - (c) $T(\mathbf{v}) = 90^\circ \text{ rotation} = (-v_2, v_1)$
 - (d) $T(\mathbf{v}) = \text{projection} = (\frac{v_1+v_2}{2}, \frac{v_1+v_2}{2})$.
- 8 Find the range and kernel (like the column space and nullspace) of T :
 - (a) $T(v_1, v_2) = (v_1 - v_2, 0)$ (b) $T(v_1, v_2, v_3) = (v_1, v_2)$
 - (c) $T(v_1, v_2) = (0, 0)$ (d) $T(v_1, v_2) = (v_1, v_1)$.
- 9 The “cyclic” transformation T is defined by $T(v_1, v_2, v_3) = (v_2, v_3, v_1)$. What is $T(T(\mathbf{v}))$? What is $T^3(\mathbf{v})$? What is $T^{100}(\mathbf{v})$? Apply T a hundred times to $\mathbf{v}$.

- 10** A linear transformation from $\mathbf{V}$ to $\mathbf{W}$ has an *inverse* from $\mathbf{W}$ to $\mathbf{V}$ when the range is all of $\mathbf{W}$ and the kernel contains only $v = \mathbf{0}$. Then $T(v) = w$ has one solution v for each w in $\mathbf{W}$. Why are these T 's not invertible?
- (a) $T(v_1, v_2) = (v_2, v_2)$ $\mathbf{W} = \mathbf{R}^2$
 (b) $T(v_1, v_2) = (v_1, v_2, v_1 + v_2)$ $\mathbf{W} = \mathbf{R}^3$
 (c) $T(v_1, v_2) = v_1$ $\mathbf{W} = \mathbf{R}^1$
- 11** If $T(v) = Av$ and A is m by n , then T is “multiplication by A .”
- (a) What are the input and output spaces $\mathbf{V}$ and $\mathbf{W}$?
 (b) Why is range of $T =$ column space of A ?
 (c) Why is kernel of $T =$ nullspace of A ?
- 12** Suppose a linear T transforms $(1, 1)$ to $(2, 2)$ and $(2, 0)$ to $(0, 0)$. Find $T(v)$:
- (a) $v = (2, 2)$ (b) $v = (3, 1)$ (c) $v = (-1, 1)$ (d) $v = (a, b)$.

Problems 13-19 may be harder. The input space $\mathbf{V}$ contains all 2 by 2 matrices M .

- 13** M is any 2 by 2 matrix and $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$. The transformation T is defined by $T(M) = AM$. What rules of matrix multiplication show that T is linear?
- 14** Suppose $A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$. Show that the range of T is the whole matrix space $\mathbf{V}$ and the kernel is the zero matrix:
- (1) If $AM = 0$ prove that M must be the zero matrix.
 (2) Find a solution to $AM = B$ for any 2 by 2 matrix B .
- 15** Suppose $A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$. Show that the identity matrix I is not in the range of T . Find a nonzero matrix M such that $T(M) = AM$ is zero.
- 16** Suppose T transposes every matrix M . Try to find a matrix A which gives $AM = M^T$ for every M . Show that no matrix A will do it. *To professors:* Is this a linear transformation that doesn't come from a matrix?
- 17** The transformation T that transposes every matrix is definitely linear. Which of these extra properties are true?
- (a) $T^2 =$ identity transformation.
 (b) The kernel of T is the zero matrix.
 (c) Every matrix is in the range of T .
 (d) $T(M) = -M$ is impossible.
- 18** Suppose $T(M) = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} M \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$. Find a matrix with $T(M) \neq 0$. Describe all matrices with $T(M) = 0$ (the kernel) and all output matrices $T(M)$ (the range).
- 19** If A and B are invertible and $T(M) = AMB$, find $T^{-1}(M)$ in the form $(\quad)M(\quad)$.

Questions 20–26 are about house transformations. The output is $T(H) = AH$.

20 How can you tell from the picture of T (house) that A is

- (a) a diagonal matrix?
- (b) a rank-one matrix?
- (c) a lower triangular matrix?

21 Draw a picture of T (house) for these matrices:

$$D = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} .7 & .7 \\ .3 & .3 \end{bmatrix} \quad \text{and} \quad U = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

22 What are the conditions on $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ to ensure that T (house) will

- (a) sit straight up?
- (b) expand the house by 3 in all directions?
- (c) rotate the house with no change in its shape?

23 Describe T (house) when $T(v) = -v + (1, 0)$. This T is “affine”.

24 Change the house matrix H to add a chimney.

25 The standard house is drawn by `plot2d(H)`. Circles from `o` and lines from `-`:

```
x = H(1,:)'; y = H(2,:)';
axis([-10 10 -10 10]), axis('square')
plot(x, y, 'o', x, y, '-');
```

Test `plot2d(A' * H)` and `plot2d(A' * A * H)` with the matrices in Figure 7.1.

26 Without a computer sketch the houses $A * H$ for these matrices A :

$$\begin{bmatrix} 1 & 0 \\ 0 & .1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} .5 & .5 \\ .5 & .5 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} .5 & .5 \\ -.5 & .5 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}.$$

27 This code creates a vector `theta` of 50 angles. It draws the unit circle and then $T(\text{circle}) = \text{ellipse}$. $T(v) = Av$ takes circles to ellipses.

```
A = [2 1; 1 2] % You can change A
theta = [0:2 * pi/50:2 * pi];
circle = [cos(theta); sin(theta)];
ellipse = A * circle;
axis([-4 4 -4 4]); axis('square')
plot(circle(1,:), circle(2,:), ellipse(1,:), ellipse(2,:))
```

28 Add two eyes and a smile to the circle in Problem 27. (If one eye is dark and the other is light, you can tell when the face is reflected across the y axis.) Multiply by matrices A to get new faces.

Challenge Problems

- 29** What conditions on $\det A = ad - bc$ ensure that the output house AH will
- (a) be squashed onto a line?
 - (b) keep its endpoints in clockwise order (not reflected)?
 - (c) have the same area as the original house?
- 30** From $A = U\Sigma V^T$ (Singular Value Decomposition) A takes **circles to ellipses**. $AV = U\Sigma$ says that the radius vectors v_1 and v_2 of the circle go to the semi-axes $\sigma_1 u_1$ and $\sigma_2 u_2$ of the ellipse. Draw the circle and the ellipse for $\theta = 30^\circ$:

$$V = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad U = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \quad \Sigma = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}.$$

- 31** Why does every linear transformation T from $\mathbf{R}^2$ to $\mathbf{R}^2$ take squares to parallelograms? Rectangles also go to parallelograms (squashed if T is not invertible).

7.2 The Matrix of a Linear Transformation

The next pages assign a matrix to every linear transformation T . For ordinary column vectors, the input v is in $V = \mathbf{R}^n$ and the output $T(v)$ is in $W = \mathbf{R}^m$. The matrix A for this transformation T will be m by n . Our choice of bases in V and W will decide A .

The standard basis vectors for $\mathbf{R}^n$ and $\mathbf{R}^m$ are the columns of I . That choice leads to a standard matrix, and $T(v) = Av$ in the normal way. But these spaces also have other bases, so the same T is represented by other matrices. A main theme of linear algebra is to choose the bases that give the best matrix for T .

When V and W are not $\mathbf{R}^n$ and $\mathbf{R}^m$, they still have bases. Each choice of basis leads to a matrix for T . When the input basis is different from the output basis, the matrix for $T(v) = v$ will not be the identity I . It will be the “change of basis matrix”.

Key idea of this section

*Suppose we know $T(v_1), \dots, T(v_n)$ for the basis vectors $v_1, \dots, v_n$.
Then linearity produces $T(v)$ for every other input vector v .*

Reason Every v is a unique combination $c_1v_1 + \dots + c_nv_n$ of the basis vectors v_i . Since T is a linear transformation (here is the moment for linearity), $T(v)$ must be the same combination $c_1T(v_1) + \dots + c_nT(v_n)$ of the known outputs $T(v_i)$.

Our first example gives the outputs $T(v)$ for the standard basis vectors $(1, 0)$ and $(0, 1)$.

Example 1 Suppose T transforms $v_1 = (1, 0)$ to $T(v_1) = (2, 3, 4)$. Suppose the second basis vector $v_2 = (0, 1)$ goes to $T(v_2) = (5, 5, 5)$. If T is linear from $\mathbf{R}^2$ to $\mathbf{R}^3$ then its “standard matrix” is 3 by 2. Those outputs $T(v_1)$ and $T(v_2)$ go into its columns:

$$A = \begin{bmatrix} 2 & 5 \\ 3 & 5 \\ 4 & 5 \end{bmatrix}. \quad \begin{array}{l} T(v_1 + v_2) = T(v_1) + T(v_2) \\ \text{combines the columns} \end{array} \quad \begin{bmatrix} 2 & 5 \\ 3 & 5 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}.$$

Example 2 The derivatives of the functions $1, x, x^2, x^3$ are $0, 1, 2x, 3x^2$. Those are four facts about the transformation T that “takes the derivative”. The inputs and the outputs are functions! Now add the crucial fact that the “derivative transformation” T is linear:

$$T(v) = \frac{dv}{dx} \quad \text{obeys the linearity rule} \quad \frac{d}{dx}(cv + dw) = c \frac{dv}{dx} + d \frac{dw}{dx}. \quad (1)$$

It is exactly this linearity that you use to find all other derivatives. From the derivative of each separate power $1, x, x^2, x^3$ (those are the basis vectors v_1, v_2, v_3, v_4) you find the derivative of any polynomial like $4 + x + x^2 + x^3$:

$$\frac{d}{dx}(4 + x + x^2 + x^3) = 1 + 2x + 3x^2 \quad (\text{because of linearity!})$$

This example applies T (the derivative d/dx) to the input $v = 4v_1 + v_2 + v_3 + v_4$. Here the input space V contains all combinations of $1, x, x^2, x^3$. I call them vectors, you might call them functions. Those four vectors are a basis for the space V of cubic polynomials (degree ≤ 3). Four derivatives tell us all derivatives in V .

For the nullspace of A , we solve $Av = 0$. For the kernel of the derivative T , we solve $dv/dx = 0$. The solution is $v = \text{constant}$. The nullspace of T is one-dimensional, containing all constant functions (like the first basis function $v_1 = 1$).

To find the range (or column space), look at all outputs from $T(v) = dv/dx$. The inputs are cubic polynomials $a + bx + cx^2 + dx^3$, so the outputs are *quadratic polynomials* (degree ≤ 2). For the output space W we have a choice. If $W = \text{cubics}$, then the range of T (the quadratics) is a subspace. If $W = \text{quadratics}$, then the range is all of W .

That second choice emphasizes the difference between the domain or input space ($V = \text{cubics}$) and the image or output space ($W = \text{quadratics}$). V has dimension $n = 4$ and W has dimension $m = 3$. The “derivative matrix” below will be 3 by 4.

The range of T is a three-dimensional subspace. The matrix will have rank $r = 3$. The kernel is one-dimensional. The sum $3 + 1 = 4$ is the dimension of the input space. This was $r + (n - r) = n$ in the Fundamental Theorem of Linear Algebra. Always *(dimension of range) + (dimension of kernel) = dimension of input space*.

Example 3 The *integral* is the inverse of the derivative. That is the Fundamental Theorem of Calculus. We see it now in linear algebra. The transformation T^{-1} that “takes the integral from 0 to x ” is linear! Apply T^{-1} to $1, x, x^2$, which are w_1, w_2, w_3 :

$$\text{Integration is } T^{-1} \quad \int_0^x 1 \, dx = x, \quad \int_0^x x \, dx = \frac{1}{2}x^2, \quad \int_0^x x^2 \, dx = \frac{1}{3}x^3.$$

By linearity, the integral of $w = B + Cx + Dx^2$ is $T^{-1}(w) = Bx + \frac{1}{2}Cx^2 + \frac{1}{3}Dx^3$. The integral of a quadratic is a cubic. The input space of T^{-1} is the quadratics, the output space is the cubics. *Integration takes W back to V* . Its matrix will be 4 by 3.

Range of T^{-1} The outputs $Bx + \frac{1}{2}Cx^2 + \frac{1}{3}Dx^3$ are cubics with no constant term.

Kernel of T^{-1} The output is zero only if $B = C = D = 0$. The nullspace is $Z = \{0\}$.

Fundamental Theorem $3 + 0$ is the dimension of the input space W for T^{-1} .

Matrices for the Derivative and Integral

We will show how the matrices A and A^{-1} copy the derivative T and the integral T^{-1} . This is an excellent example from calculus. (I write A^{-1} but I don't quite mean it.) Then comes the general rule—how to represent any linear transformation T by a matrix A .

The derivative transforms the space V of cubics to the space W of quadratics. The basis for V is $1, x, x^2, x^3$. The basis for W is $1, x, x^2$. *The derivative matrix is 3 by 4:*

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix} = \text{matrix form of derivative } T. \quad (2)$$

Why is A the correct matrix? Because ***multiplying by A agrees with transforming by T*** . The derivative of $v = a + bx + cx^2 + dx^3$ is $T(v) = b + 2cx + 3dx^2$. The same numbers b and $2c$ and $3d$ appear when we multiply by the matrix A :

$$\text{Take the derivative} \quad \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} b \\ 2c \\ 3d \end{bmatrix}. \quad (3)$$

Look also at T^{-1} . The integration matrix is 4 by 3. Watch how the following matrix starts with $w = B + Cx + Dx^2$ and produces its integral $0 + Bx + \frac{1}{2}Cx^2 + \frac{1}{3}Dx^3$:

$$\text{Take the integral} \quad \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{3} \end{bmatrix} \begin{bmatrix} B \\ C \\ D \end{bmatrix} = \begin{bmatrix} 0 \\ B \\ \frac{1}{2}C \\ \frac{1}{3}D \end{bmatrix}. \quad (4)$$

I want to call that matrix A^{-1} , and I will. But you realize that rectangular matrices don't have inverses. At least they don't have two-sided inverses. This rectangular A has a *one-sided inverse*. The integral is a one-sided inverse of the derivative!

$$AA^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{but} \quad A^{-1}A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

If you integrate a function and then differentiate, you get back to the start. So $AA^{-1} = I$. But if you differentiate before integrating, the constant term is lost. ***The integral of the derivative of 1 is zero:***

$$T^{-1}T(1) = \text{integral of zero function} = 0.$$

This matches $A^{-1}A$, whose first column is all zero. The derivative T has a kernel (the constant functions). Its matrix A has a nullspace. Main point again: Av copies $T(v)$.

Construction of the Matrix

Now we construct a matrix for any linear transformation. Suppose T transforms the space V (n -dimensional) to the space W (m -dimensional). We choose a basis $v_1, \dots, v_n$ for V and we choose a basis $w_1, \dots, w_m$ for W . The matrix A will be m by n . To find the first column of A , apply T to the first basis vector v_1 . The output $T(v_1)$ is in W .

$T(v_1)$ is a combination $a_{11}w_1 + \dots + a_{m1}w_m$ of the output basis for W .

These numbers $a_{11}, \dots, a_{m1}$ go into the first column of A . Transforming v_1 to $T(v_1)$ matches multiplying $(1, 0, \dots, 0)$ by A . It yields that first column of the matrix.

When T is the derivative and the first basis vector is 1, its derivative is $T(v_1) = 0$. So for the derivative matrix, the first column of A was all zero.

For the integral, the first basis function is again 1. Its integral is the second basis function x . So the first column of A^{-1} was $(0, 1, 0, 0)$. Here is the construction of A .

Key rule: The j th column of A is found by applying T to the j th basis vector v_j

$$T(v_j) = \text{combination of basis vectors of } \mathbf{W} = a_{1j}w_1 + \cdots + a_{mj}w_m. \quad (5)$$

These numbers $a_{1j}, \dots, a_{mj}$ go into column j of A . *The matrix is constructed to get the basis vectors right. Then linearity gets all other vectors right.* Every v is a combination $c_1v_1 + \cdots + c_nv_n$, and $T(v)$ is a combination of the w 's. When A multiplies the coefficient vector $c = (c_1, \dots, c_n)$ in the v combination, Ac produces the coefficients in the $T(v)$ combination. This is because matrix multiplication (combining columns) is linear like T .

The matrix A tells us what T does. Every linear transformation from $\mathbf{V}$ to $\mathbf{W}$ can be converted to a matrix. This matrix depends on the bases.

Example 4 *If the bases change, T is the same but the matrix A is different.*

Suppose we reorder the basis to $x, x^2, x^3, 1$ for the cubics in $\mathbf{V}$. Keep the original basis $1, x, x^2$ for the quadratics in $\mathbf{W}$. The derivative of the first basis vector $v_1 = x$ is the first basis vector $w_1 = 1$. So the first column of A looks different:

$$A_{\text{new}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \end{bmatrix} = \begin{array}{l} \text{matrix for the derivative } T \\ \text{when the bases change to} \\ x, x^2, x^3, 1 \text{ and } 1, x, x^2. \end{array}$$

When we reorder the basis of $\mathbf{V}$, we reorder the columns of A . The input basis vector v_j is responsible for column j . The output basis vector w_i is responsible for row i . Soon the changes in the bases will be more than permutations.

Products AB Match Transformations TS

The examples of derivative and integral made three points. First, linear transformations T are everywhere—in calculus and differential equations and linear algebra. Second, spaces other than $\mathbf{R}^n$ are important—we had functions in $\mathbf{V}$ and $\mathbf{W}$. Third, T still boils down to a matrix A . Now we make sure that we can find this matrix.

The next examples have $\mathbf{V} = \mathbf{W}$. We choose the same basis for both spaces. Then we can compare the matrices A^2 and AB with the transformations T^2 and TS .

Example 5 T rotates every vector by the angle θ . Here $\mathbf{V} = \mathbf{W} = \mathbf{R}^2$. Find A .

Solution The standard basis is $v_1 = (1, 0)$ and $v_2 = (0, 1)$. To find A , apply T to those basis vectors. In Figure 7.3a, they are rotated by θ . *The first vector $(1, 0)$ swings around to $(\cos \theta, \sin \theta)$.* This equals $\cos \theta$ times $(1, 0)$ plus $\sin \theta$ times $(0, 1)$. Therefore those

numbers $\cos \theta$ and $\sin \theta$ go into the *first column* of A :

$$\begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix} \text{ shows column 1} \quad A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \text{ shows both columns.}$$

For the second column, transform the second vector $(0, 1)$. The figure shows it rotated to $(-\sin \theta, \cos \theta)$. *Those numbers go into the second column.* Multiplying A times $(0, 1)$ produces that column. A agrees with T on the basis, and on all v .

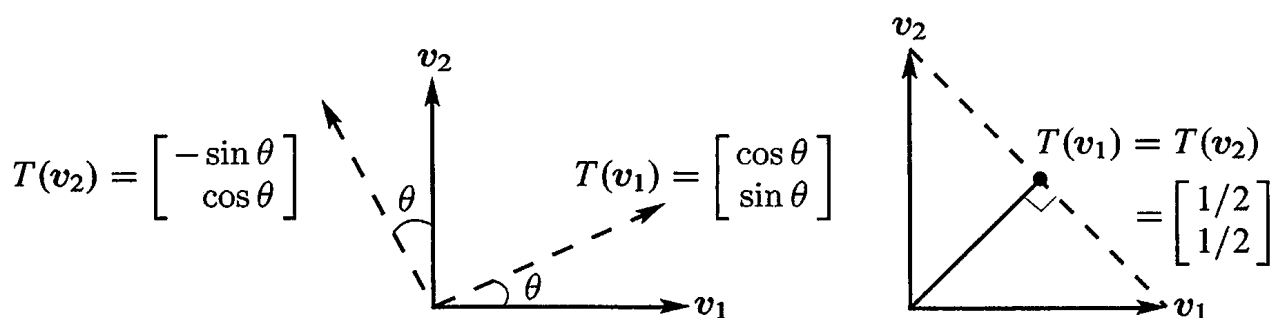


Figure 7.3: Two transformations: Rotation by θ and projection onto the 45° line.

Example 6 (*Projection*) Suppose T projects every plane vector onto the 45° line. Find its matrix for two different choices of the basis. We will find two matrices.

Solution Start with a specially chosen basis, not drawn in Figure 7.3. The basis vector v_1 is along the 45° line. *It projects to itself:* $T(v_1) = v_1$. So the first column of A contains 1 and 0. The second basis vector v_2 is along the perpendicular line (135°). *This basis vector projects to zero.* So the second column of A contains 0 and 0:

$$\text{Projection} \quad A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{when } \mathbf{V} \text{ and } \mathbf{W} \text{ have the } 45^\circ \text{ and } 135^\circ \text{ basis.}$$

Now take the standard basis $(1, 0)$ and $(0, 1)$. Figure 7.3b shows how $(1, 0)$ projects to $(\frac{1}{2}, \frac{1}{2})$. That gives the first column of A . The other basis vector $(0, 1)$ also projects to $(\frac{1}{2}, \frac{1}{2})$. So the standard matrix for this projection is A :

$$\text{Same projection} \quad A = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \quad \text{for the standard basis.}$$

Both A 's are projection matrices. If you square A it doesn't change. Projecting twice is the same as projecting once: $T^2 = T$ so $A^2 = A$. Notice what is hidden in that statement: *The matrix for T^2 is A^2 .*

We have come to something important—the real reason for the way matrices are multiplied. *At last we discover why!* Two transformations S and T are represented by two matrices B and A . When we apply T to the output from S , we get the “composition” TS . When we apply A after B , we get the matrix product AB . **Matrix multiplication gives the correct matrix AB to represent TS .**

The transformation S is from a space U to V . Its matrix B uses a basis $u_1, \dots, u_p$ for U and a basis $v_1, \dots, v_n$ for V . The matrix is n by p . The transformation T is from V to W as before. *Its matrix A must use the same basis $v_1, \dots, v_n$ for V —this is the output space for S and the input space for T . Then the matrix AB matches TS :*

Multiplication The linear transformation TS starts with any vector u in U , goes to $S(u)$ in V and then to $T(S(u))$ in W . The matrix AB starts with any x in $\mathbb{R}^p$, goes to Bx in $\mathbb{R}^n$ and then to ABx in $\mathbb{R}^m$. The matrix AB correctly represents TS :

$$TS : U \rightarrow V \rightarrow W \quad AB : (m \text{ by } n)(n \text{ by } p) = (m \text{ by } p).$$

The input is $u = x_1 u_1 + \dots + x_p u_p$. The output $T(S(u))$ matches the output ABx . **Product of transformations matches product of matrices.**

The most important cases are when the spaces U, V, W are the same and their bases are the same. With $m = n = p$ we have square matrices.

Example 7 S rotates the plane by θ and T also rotates by θ . Then TS rotates by 2θ . This transformation T^2 corresponds to the rotation matrix A^2 through 2θ :

$$T = S \quad A = B \quad T^2 = \text{rotation by } 2\theta \quad A^2 = \begin{bmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{bmatrix}. \quad (6)$$

By matching (transformation)² with (matrix)², we pick up the formulas for $\cos 2\theta$ and $\sin 2\theta$. Multiply A times A :

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} \cos^2 \theta - \sin^2 \theta & -2 \sin \theta \cos \theta \\ 2 \sin \theta \cos \theta & \cos^2 \theta - \sin^2 \theta \end{bmatrix}. \quad (7)$$

Comparing (6) with (7) produces $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$ and $\sin 2\theta = 2 \sin \theta \cos \theta$. Trigonometry (the double angle rule) comes from linear algebra.

Example 8 S rotates by θ and T rotates by $-\theta$. Then $TS = I$ matches $AB = I$.

In this case $T(S(u))$ is u . We rotate forward and back. For the matrices to match, ABx must be x . The two matrices are inverses. Check this by putting $\cos(-\theta) = \cos \theta$ and $\sin(-\theta) = -\sin \theta$ into the backward rotation matrix:

$$AB = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} \cos^2 \theta + \sin^2 \theta & 0 \\ 0 & \cos^2 \theta + \sin^2 \theta \end{bmatrix} = I.$$